

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 227 factors, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

■ Long
 ■ Short
 ■ Neutral

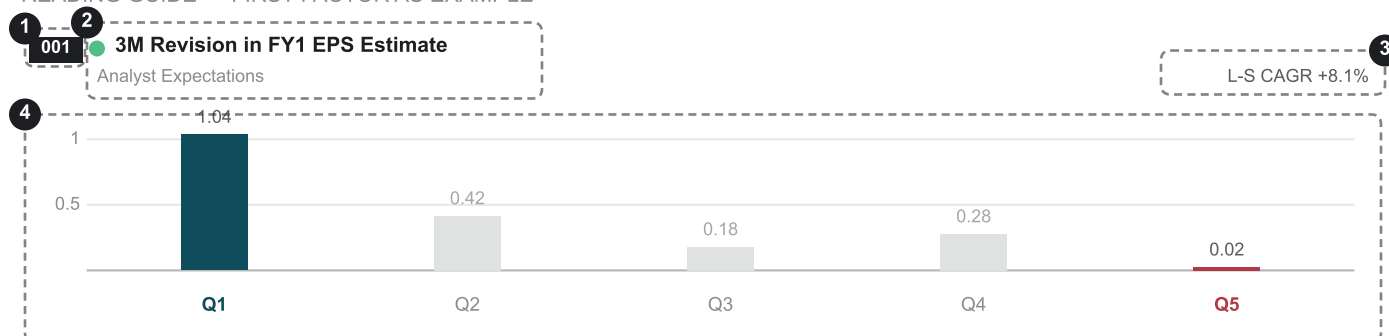
Long / short quintiles follow the live 48-month rules — spread-threshold based, not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

FACTOR STYLES

● Valuation	35	● Price Momentum	50
● Earnings Quality	47	● Size	4
● Analyst Expectations	18	● Historical Growth	38
● Capital Efficiency	27	● Volatility	8

001 In current model portfolio (29 factors)

READING GUIDE — FIRST FACTOR AS EXAMPLE

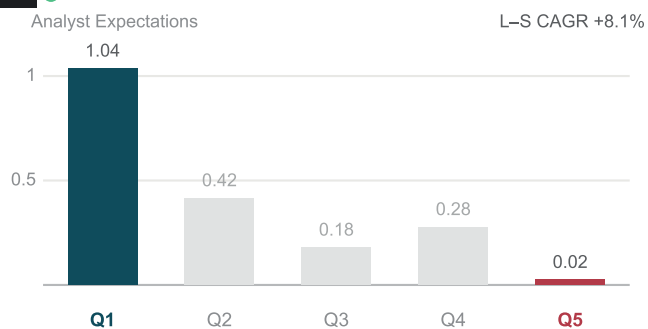
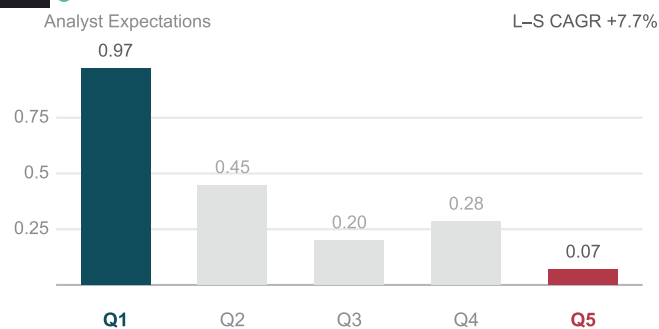
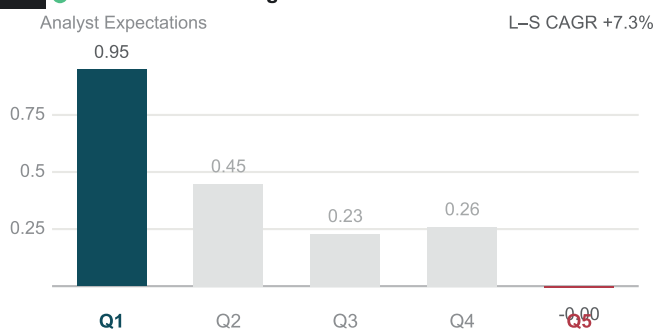
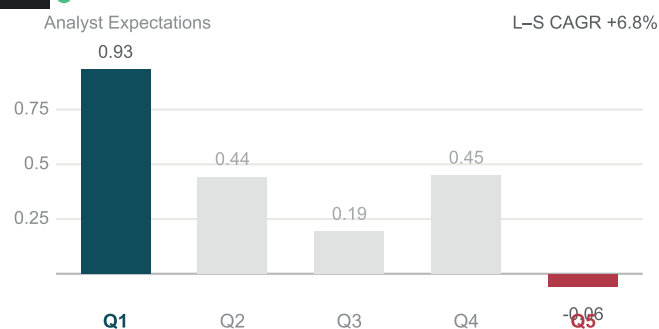
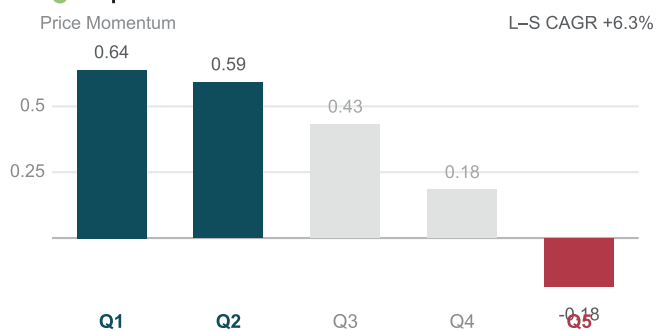
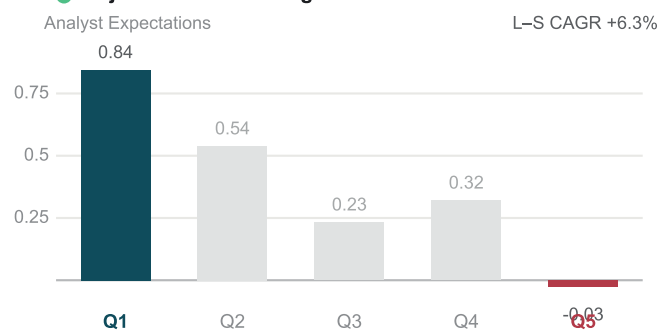


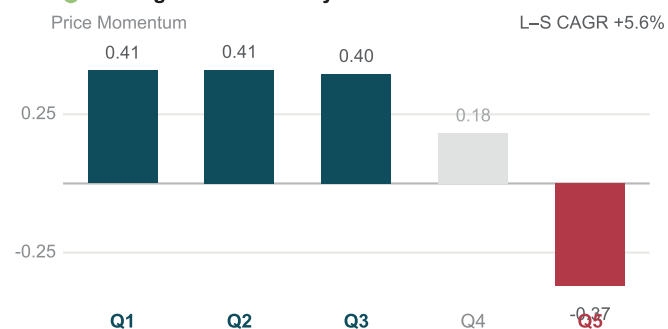
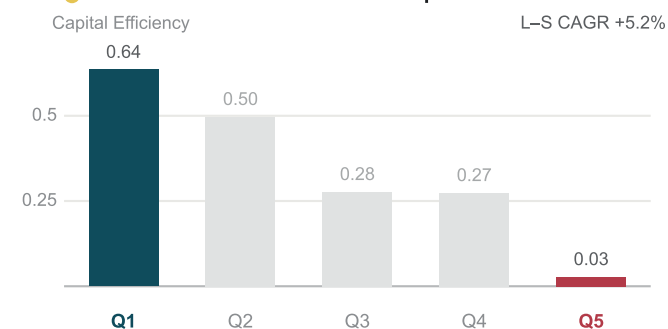
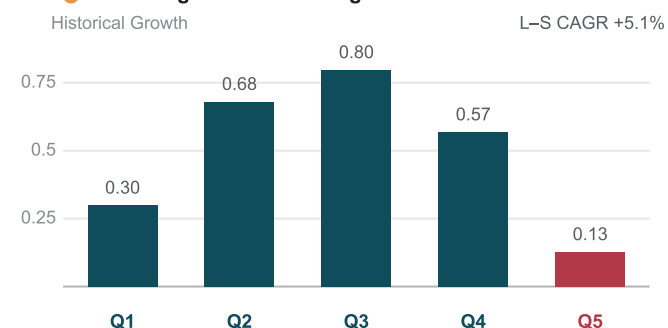
1 Rank - ordered by **3** L-S CAGR, descending.
Filled dark badge = in the current Top-50 portfolio.

2 Factor name - colour dot = style (legend above).

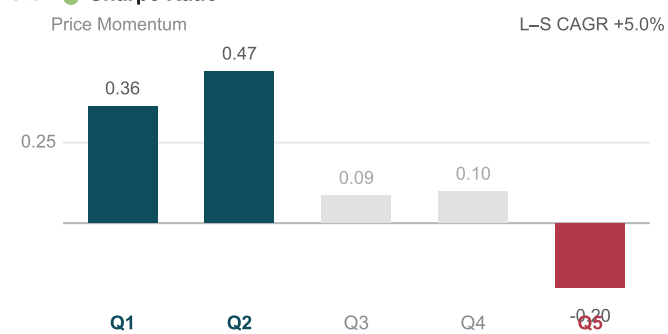
3 L-S CAGR - LAST 48 MONTHS only (rolling IS), net of 20 bp cost. The live selection / sort basis. The L/S labels and sector exclusions shown are these same live 48M rules.

4 Bars - FULL HISTORY (Jul 2015 - Aug 2026): avg monthly return of the five quintiles, BEFORE costs. L/S tags are the live 48M rules; only the return window differs from **3** (so signs can disagree).

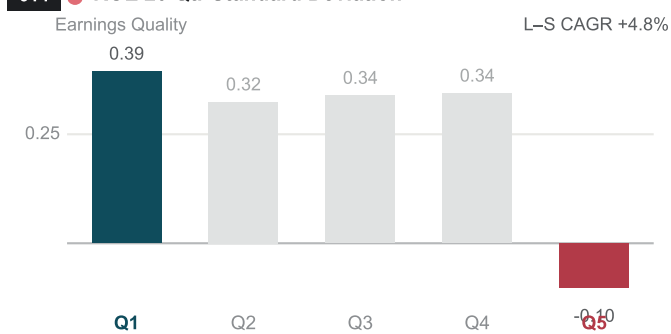
001 ● 3M Revision in FY1 EPS Estimate**002 ● 3M Revision in FY2 EPS Estimate****003 ● Street Revision Magnitude****004 ● Number of EPS FY1 Revisions****005 ● Slope of 52W Price Trend****006 ● Adjusted Revision Magnitude**

007 ● **6M Chg in Target Price**008 ● **Standardized Unexpected Earnings**009 ● **Closing Price to 260 Day Low**010 ● **Cash Flow Return on Invested Capital**011 ● **3Y Average Annual Earnings Growth**012 ● **3Y Chg in Earnings per Share**

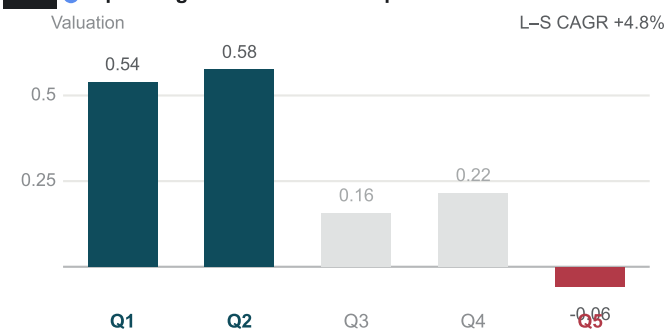
013 ● Sharpe Ratio



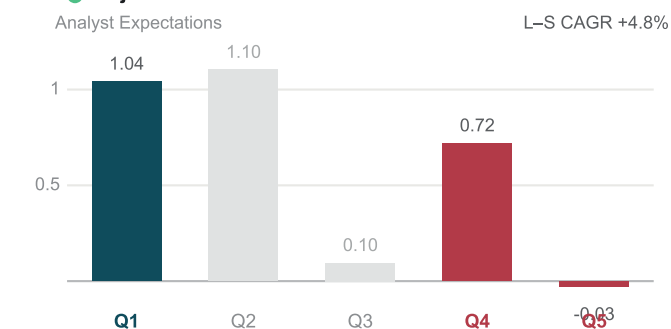
014 ● ROE 20 Qtr Standard Deviation



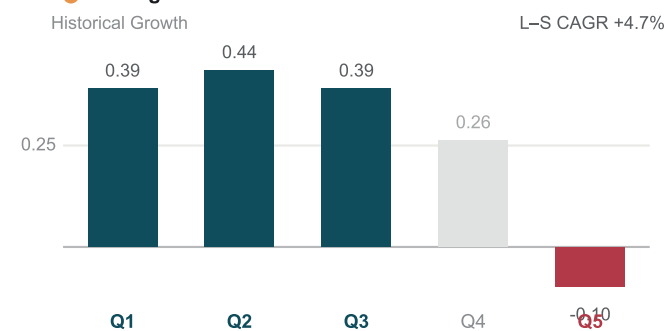
015 ● Operating Cash Flow to Enterprise Value



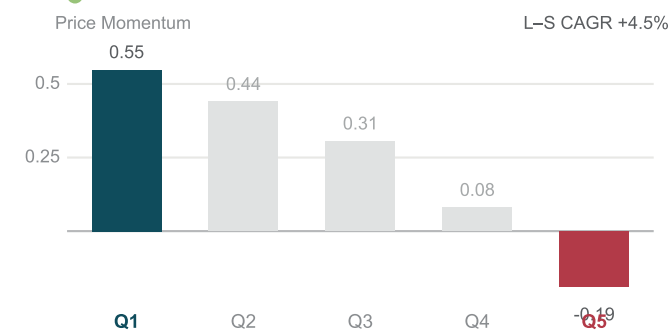
016 ● Adjusted Number of EPS FY1 Revisions

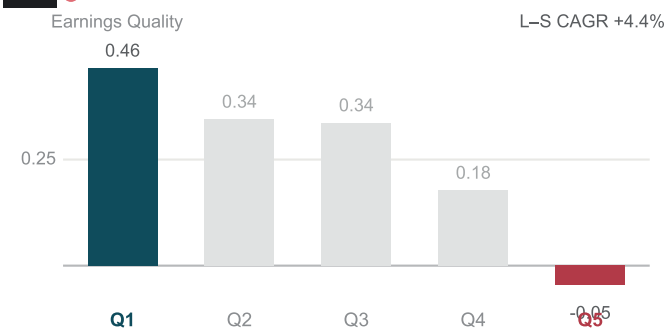
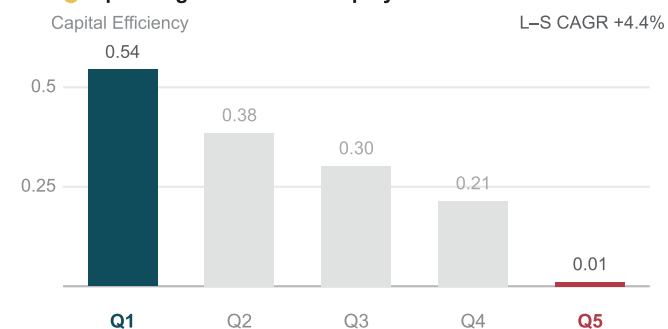
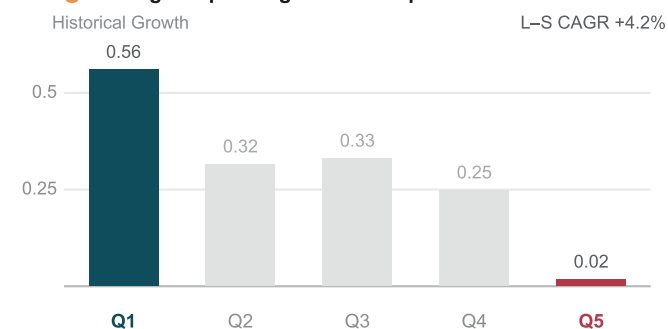
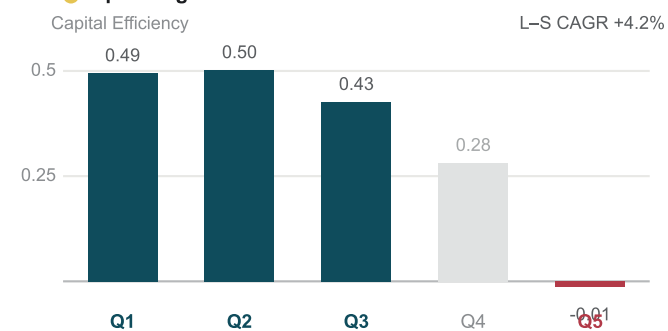
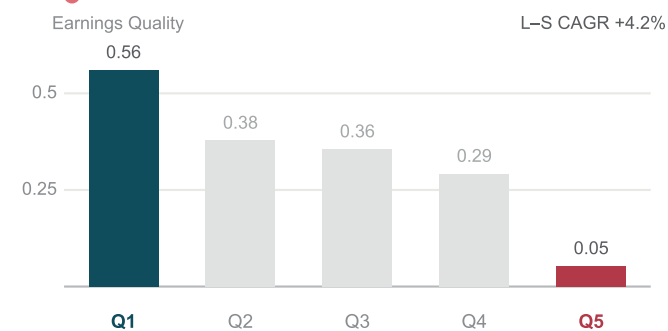


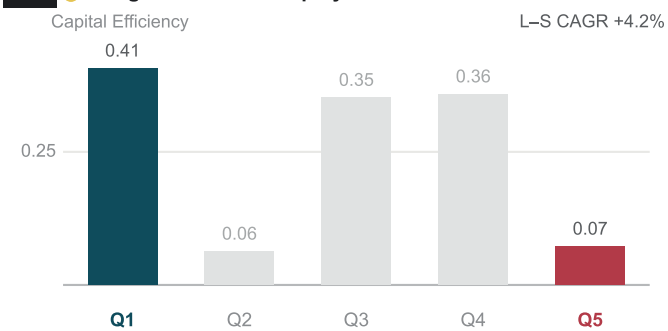
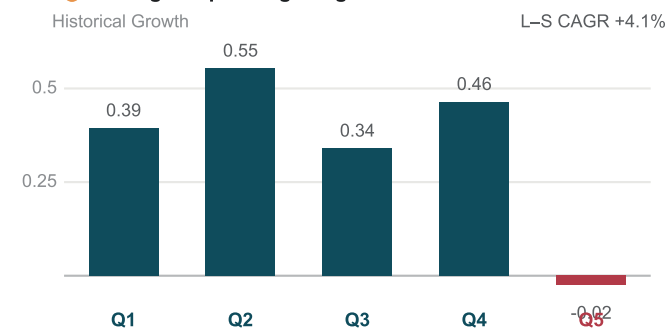
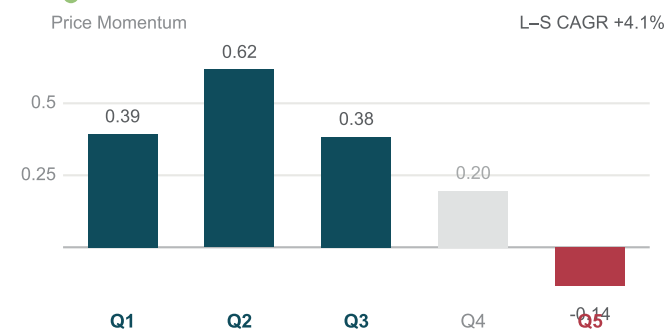
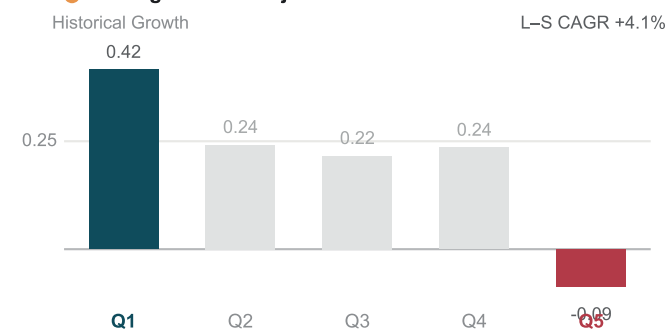
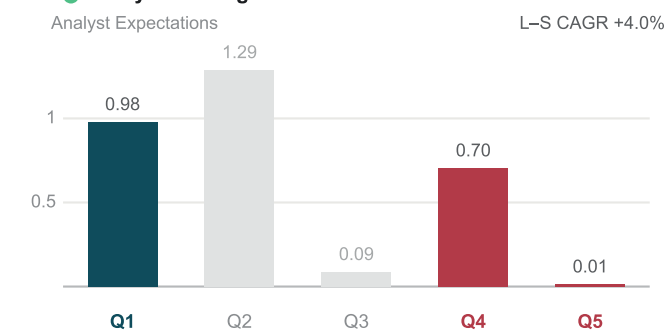
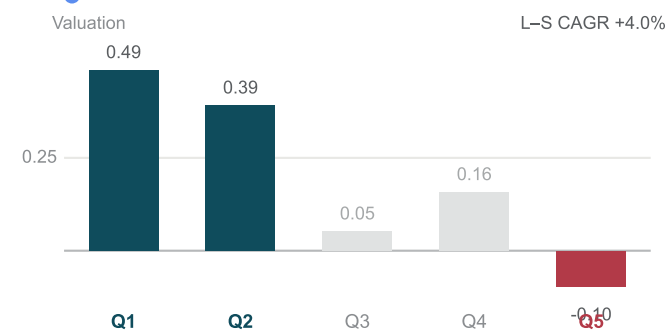
017 ● 1Y Chg in Sales Turnover

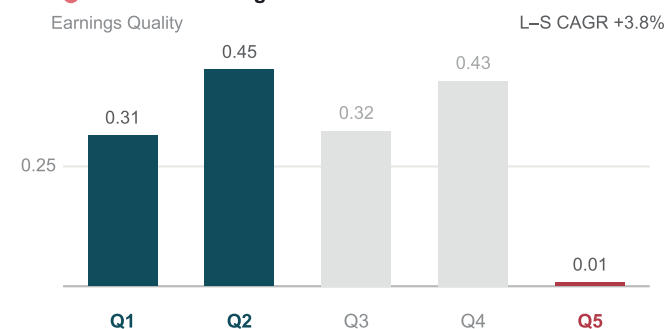
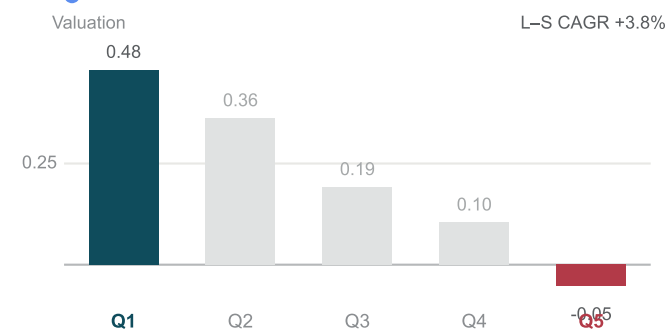
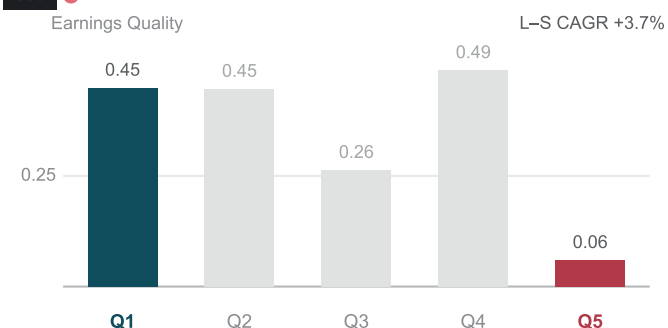


018 ● 24M Residual Return Variance

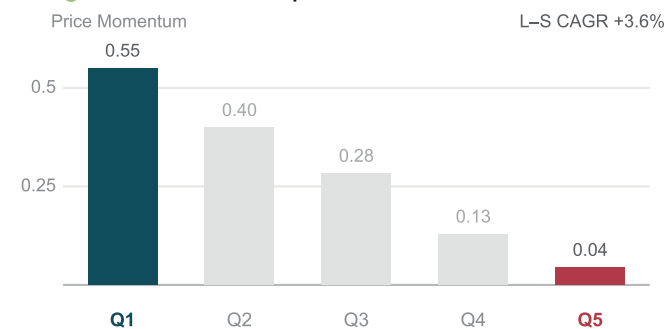


019 ROA 20 Qtr Standard Deviation**020 Net Profit Margin****021 Operating Cash Flow to Equity****022 1Y Chg in Operating Cash Flow per Share****023 Operating Cash Flow Ratio****024 Free Cash Flow to Sales**

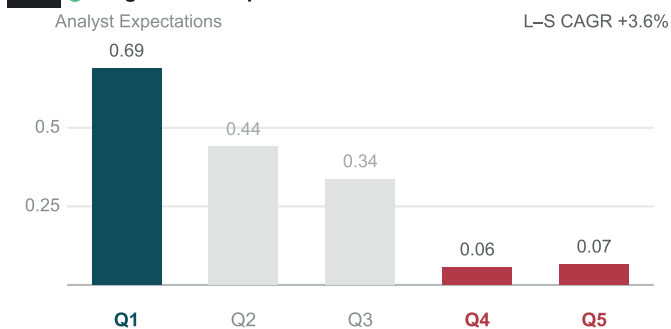
025 ● Long Term Debt to Equity Ratio**026 ● 3Y Chg in Operating Margin****027 ● 180D Price TStat****028 ● 1Y Chg in Asset Adjusted Free Cash Flow****029 ● Analyst Earnings Estimate Diffusion****030 ● Sales to Price Ratio**

031 ● **Free Cash Flow to Common Equity**032 ● **EBIT Margin**033 ● **Gross Profit Margin**034 ● **Sales to EV Ratio**035 ● **Asset Turnover**036 ● **6M Avg Chg 1M Recommendation**

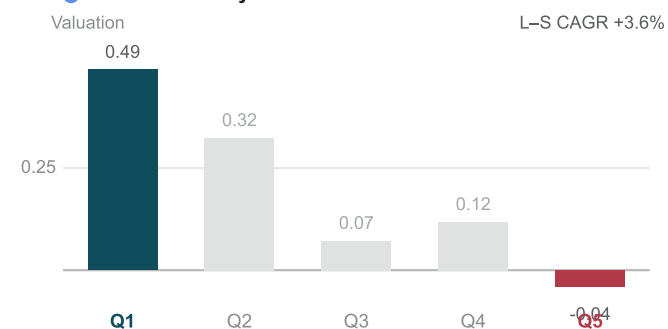
037 ● Institution Ownership Concentration



038 ● Target Price Gap to 6M EMA



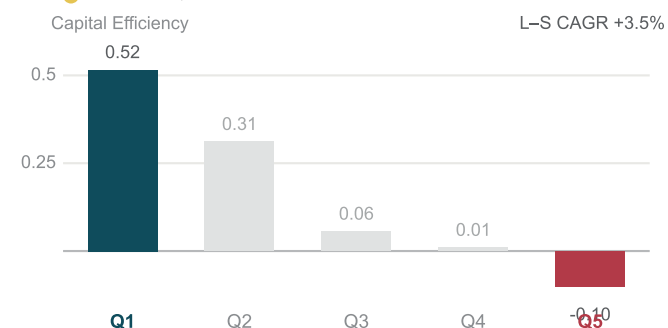
039 ● Current Liability to Price



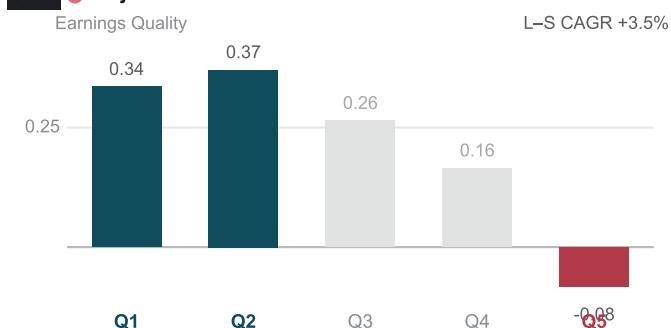
040 ● Operating Cash Flow to Price



041 ● Tobin's Q Ratio



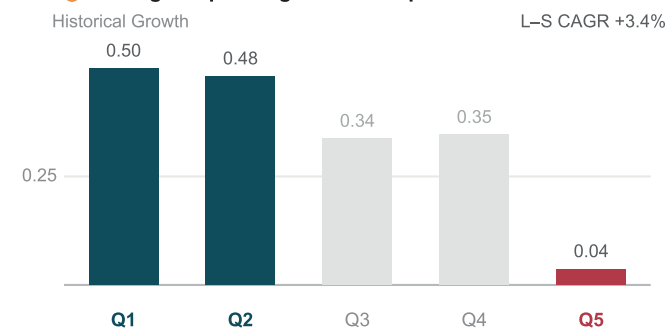
042 ● Adjusted Accruals



043 ● 1Y Chg in Sales to Price



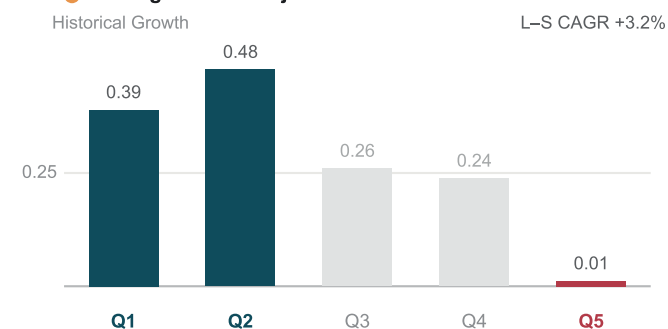
044 ● 3Y Chg in Operating Cash Flow per Share



045 ● 1Y Chg in Free Cash Flow per Share



046 ● 1Y Chg in Asset Adjusted EPS

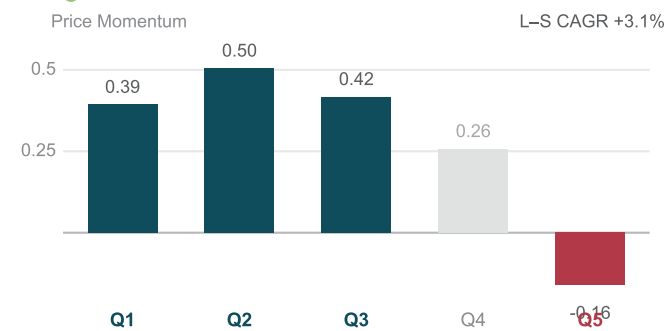
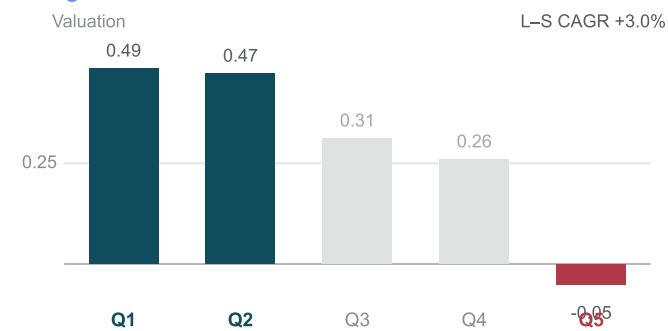
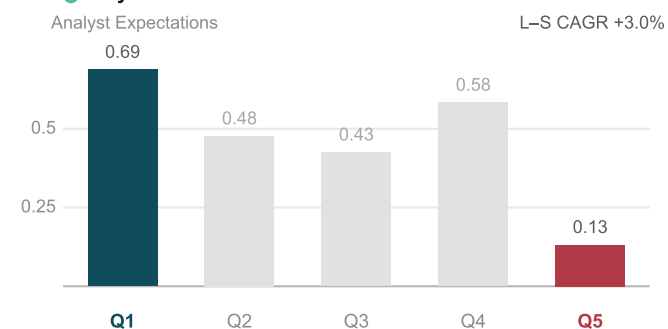
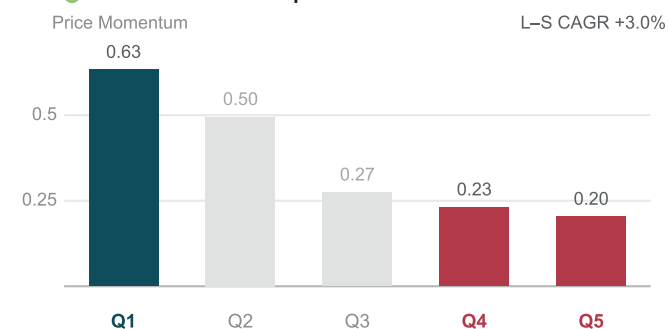
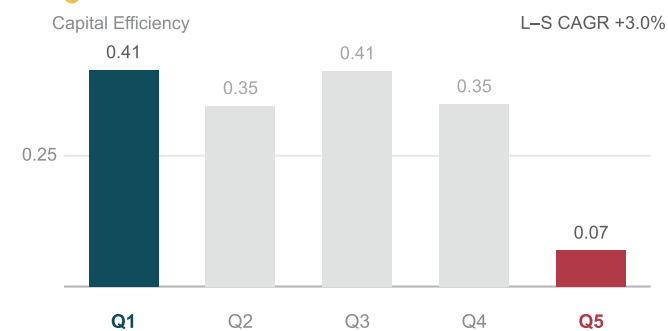


047 ● 3Y Chg in Free Cash Flow per Share

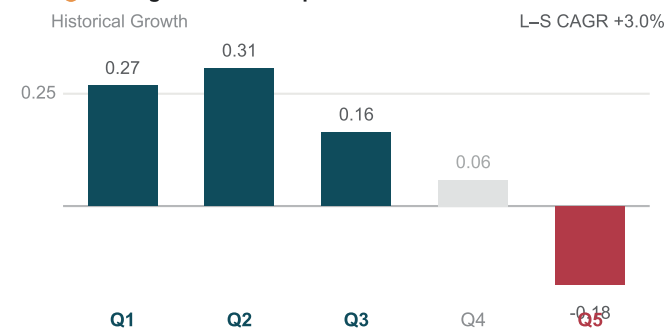


048 ● Capital Acquisition Ratio

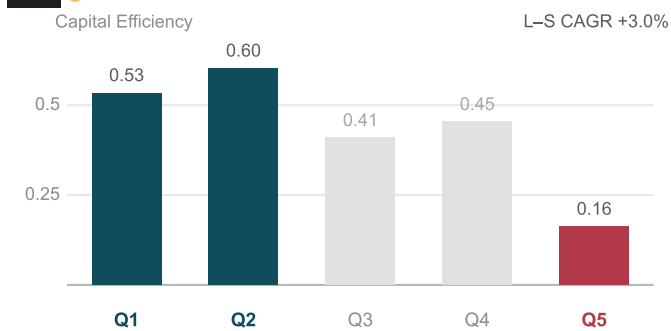


049 ● **Share Turnover**050 ● **Growth Flow to Price**051 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**052 ● **Institution Ownership Level**053 ● **1Y Chg in Operating Margin**054 ● **Gross Profit to Assets**

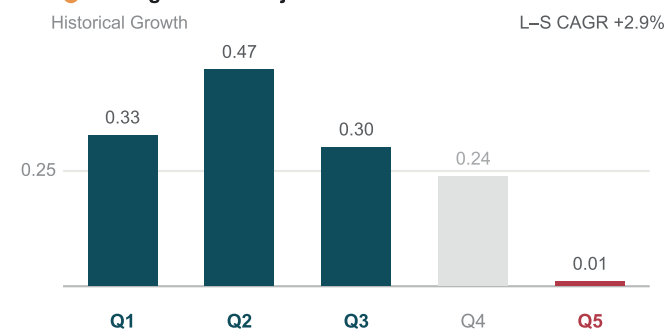
055 ● 1Y Chg in Cash Flow per Share



056 ● Receivables Turnover Ratio



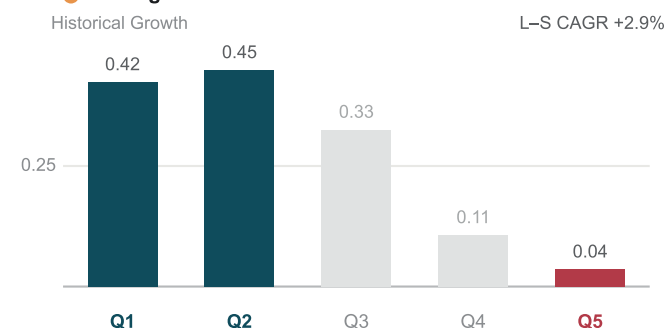
057 ● 1Y Chg in Asset Adjusted Cash Flow



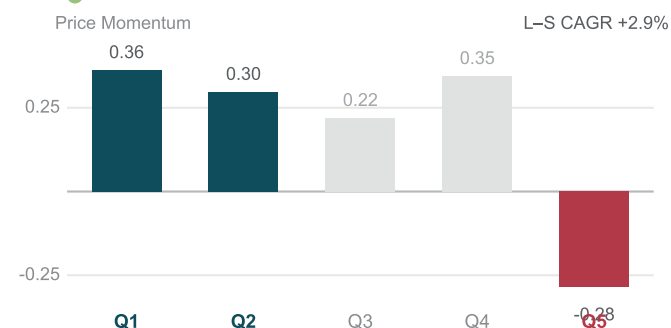
058 ● 3Y Chg in Sales Turnover

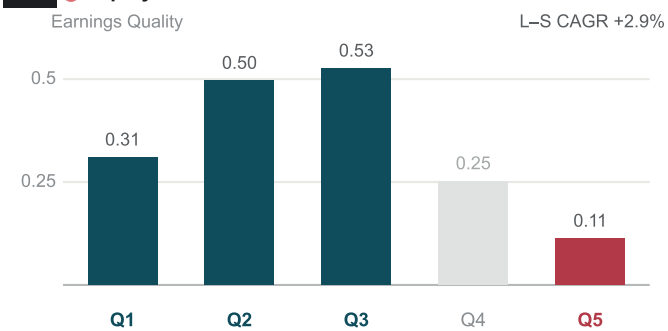
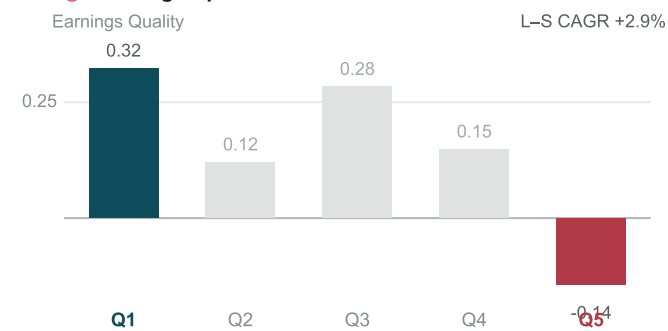
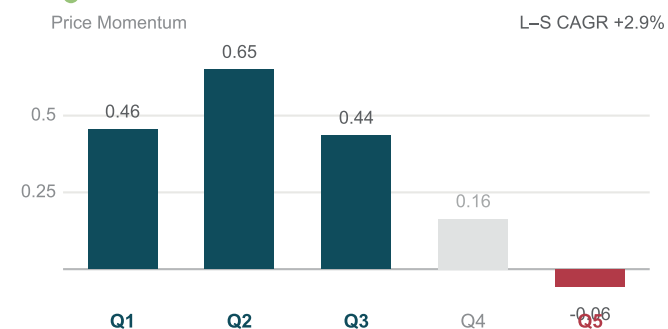
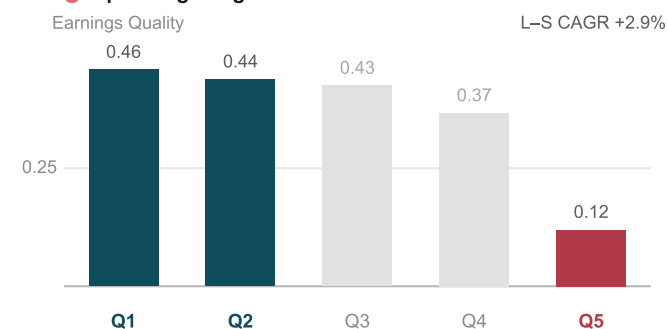
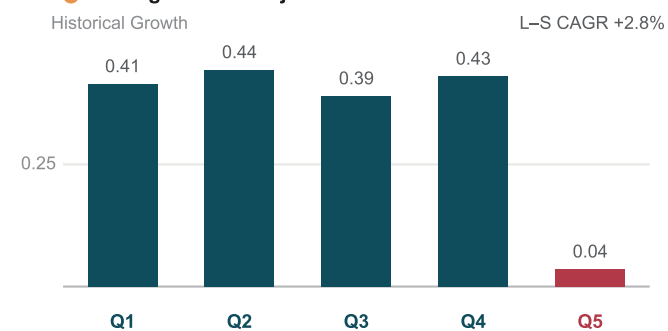
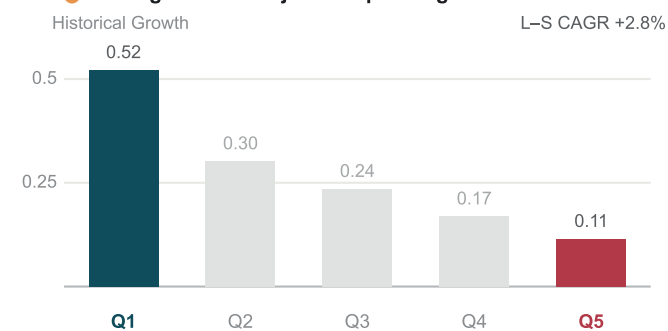


059 ● 1Y Chg in Cash Flow to Price

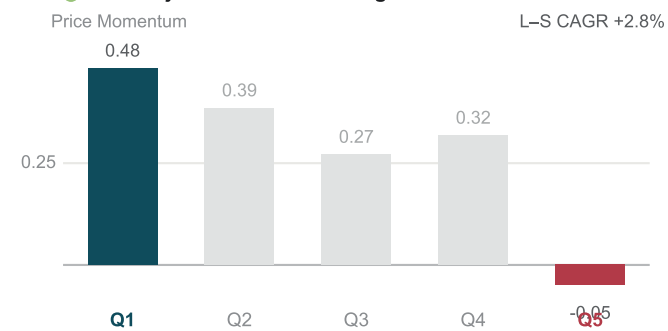


060 ● 6 Months Price Reversal

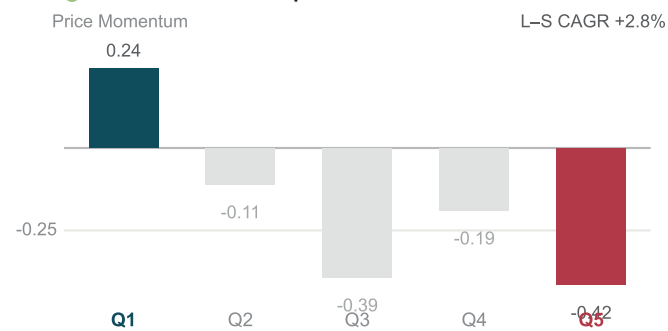


061 Equity Turnover**062 Working Capital Accruals****063 15/36 Week Price Ratio****064 Operating Margin****065 3Y Chg in Asset Adjusted Cash Flow****066 1Y Chg in Asset Adjusted Operating Cash Flow**

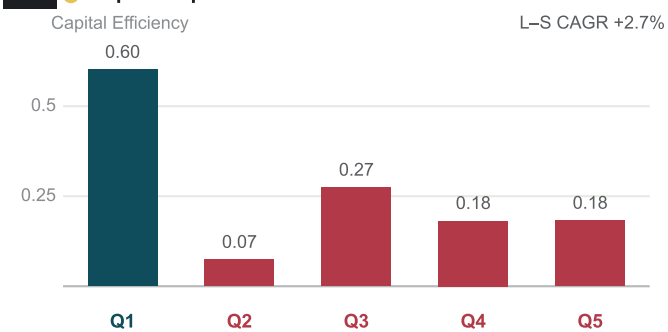
067 ● Risk Adjusted Relative Strength



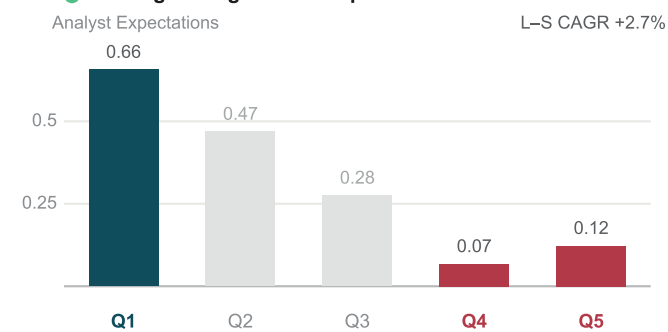
068 ● Institution Ownership Turnover



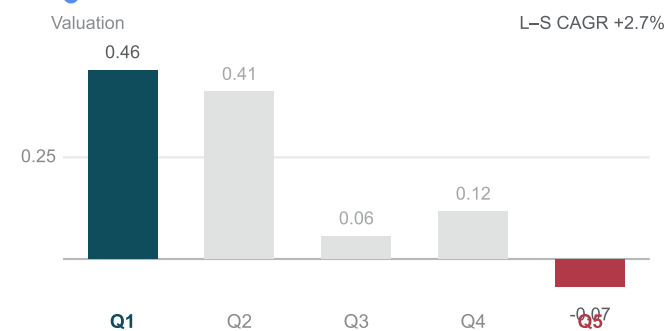
069 ● Capital Expenditure to Total Assets



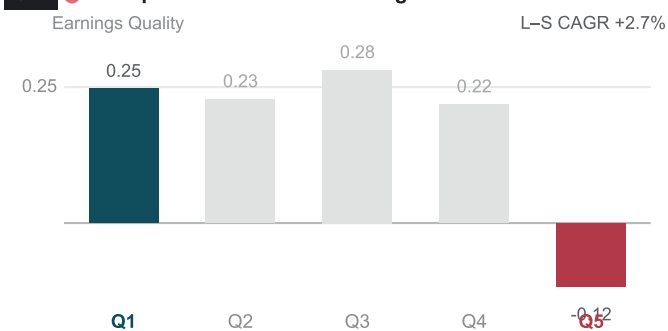
070 ● 6M Chg in Target Price Gap



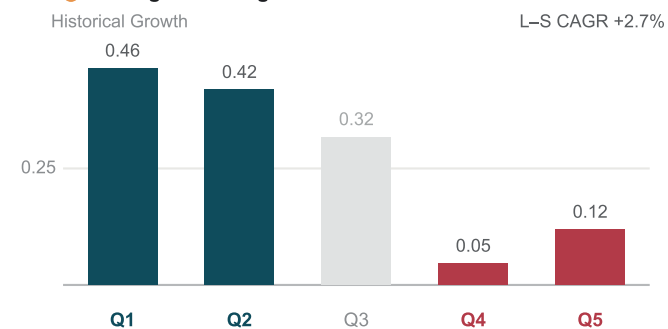
071 ● Assets to Price Ratio



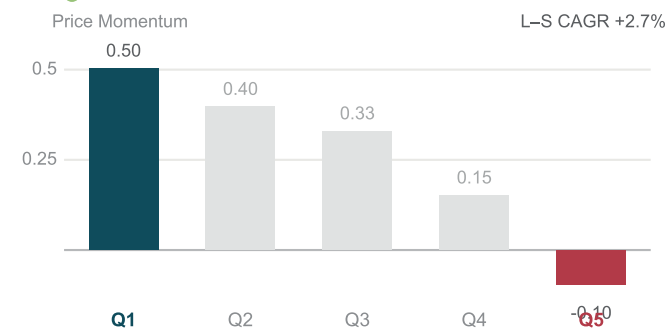
072 ● Unexpected Receivables Change



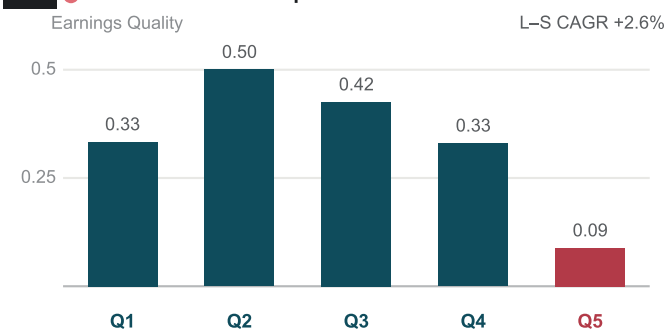
073 ● 1Y Chg in Earnings to Price



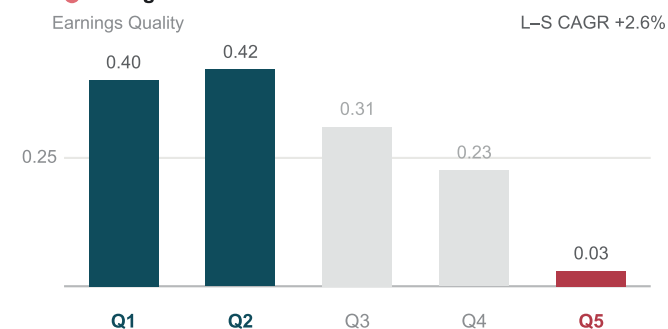
074 ● 12M - 1M Price Momentum



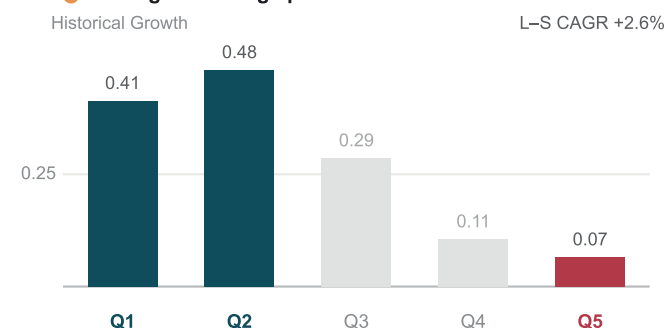
075 ● Sales to Invested Capital



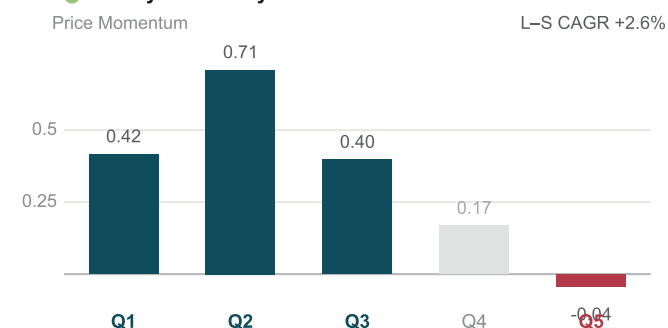
076 ● Change EPS to Sales



077 ● 1Y Chg in Earnings per Share

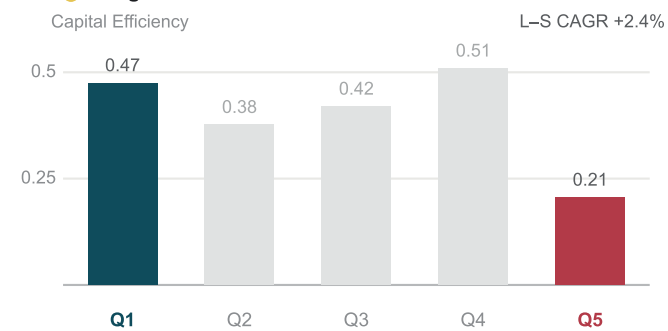


078 ● 50 Day to 200 Day Stock Price

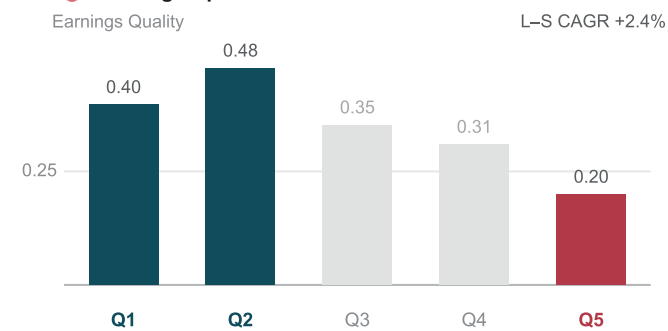




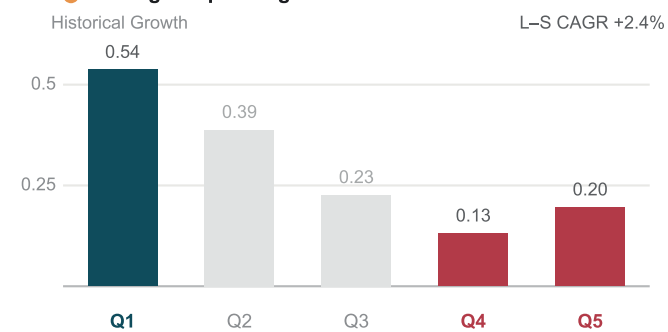
085 ● Long Term Debt to Assets Ratio



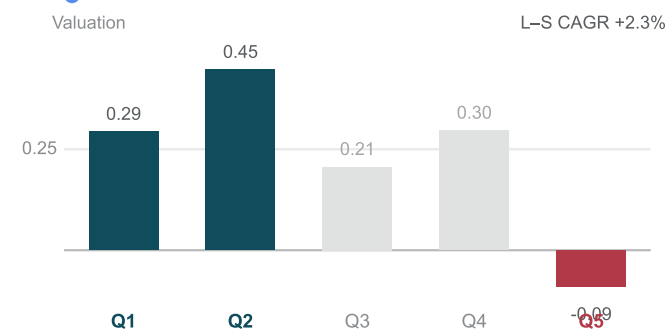
086 ● Working Capital to Assets Ratio



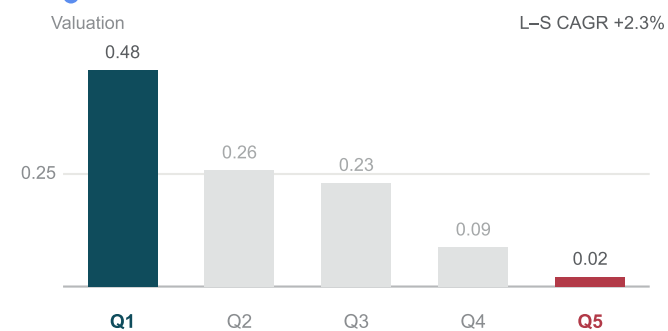
087 ● 3Y Chg in Operating Cash Flow to Price



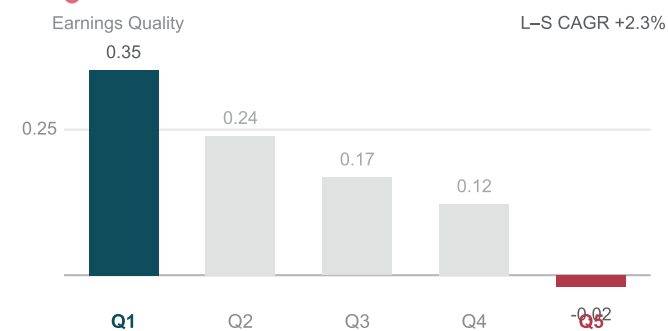
088 ● Dividends to Price Ratio

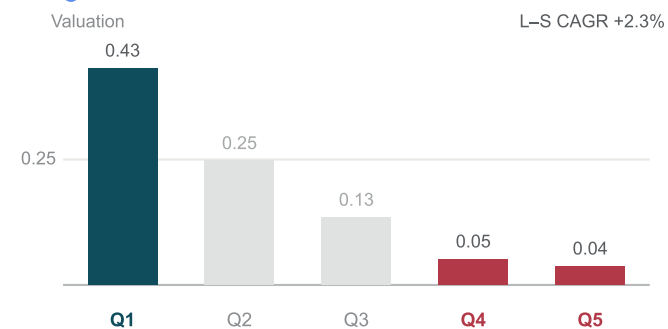
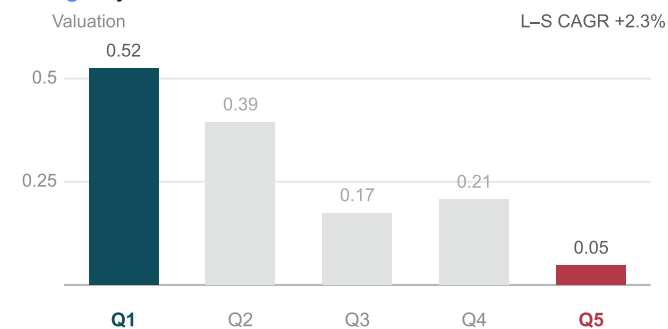
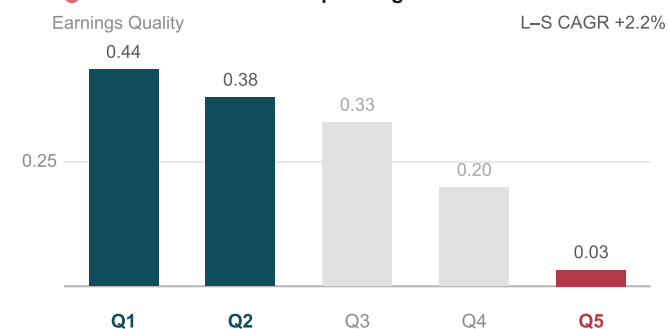
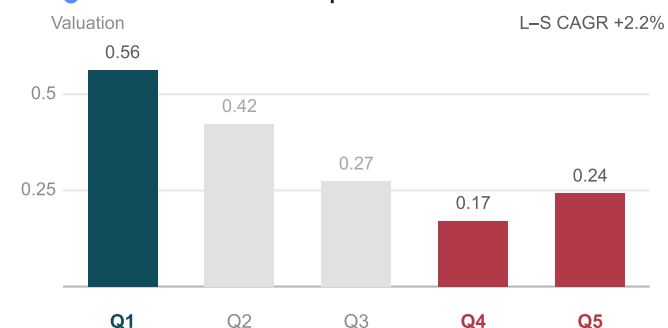
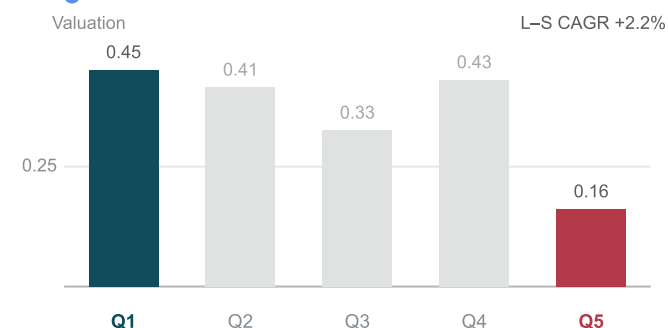


089 ● Cash to Price

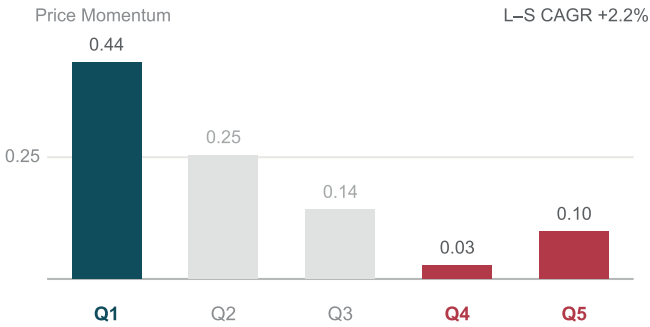


090 ● Accrual Ratio - Balance Sheet

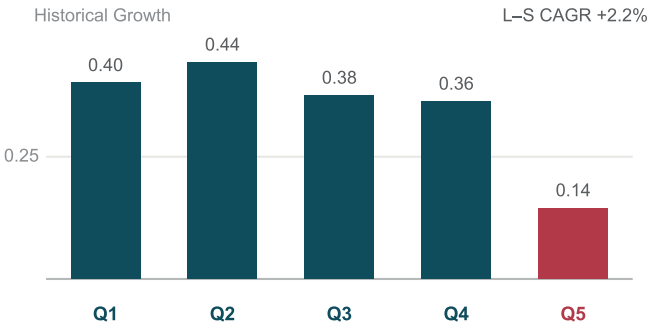


091 ● **Cash to EV Ratio**092 ● **Adjusted EBIT to Price**093 ● **Depreciation to Capex Ratio**094 ● **Pretax Income to Net Operating Assets**095 ● **Free Cash Flow to Enterprise Value**096 ● **Cash to Total Assets**

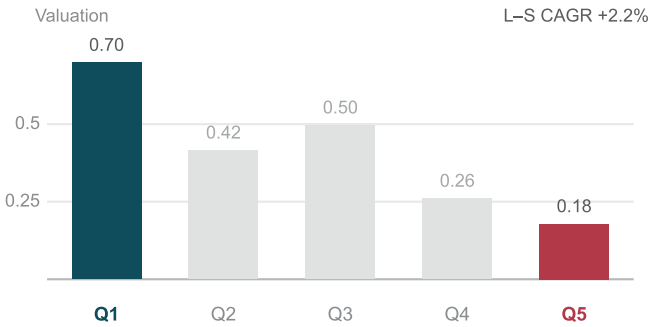
097 ● Log of Unadjusted Stock Price



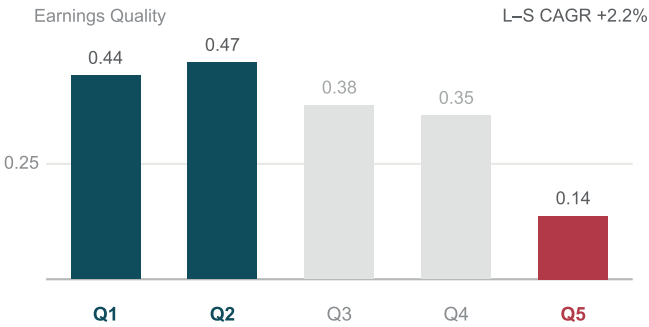
098 ● 3Y Chg in Asset Adjusted EPS



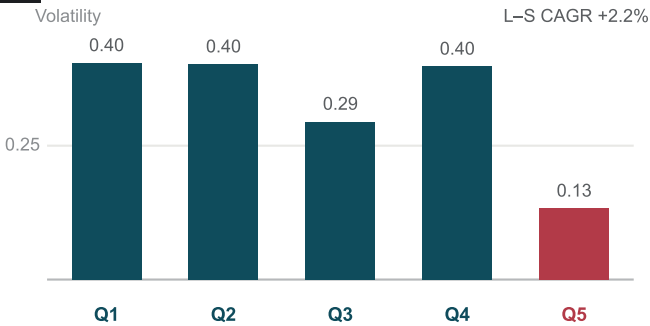
099 ● Forward Earnings to Price



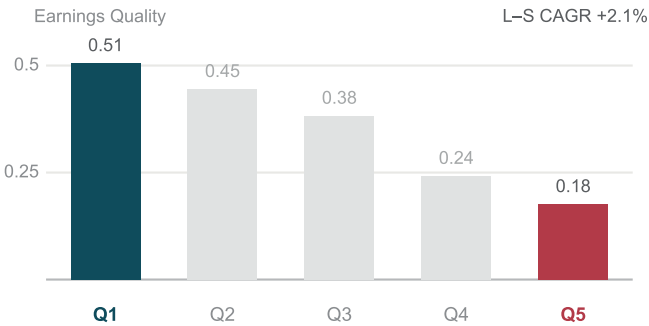
100 ● Cash Flow to Total Asset



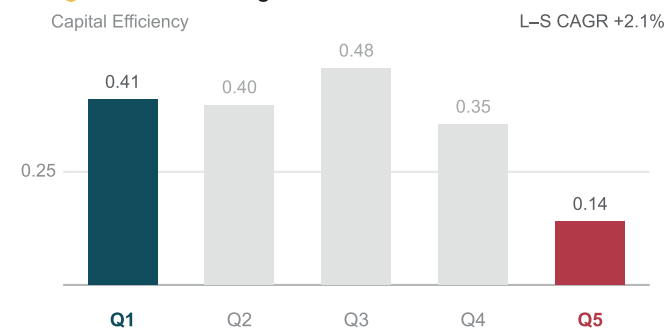
101 ● 60M CAPM Beta



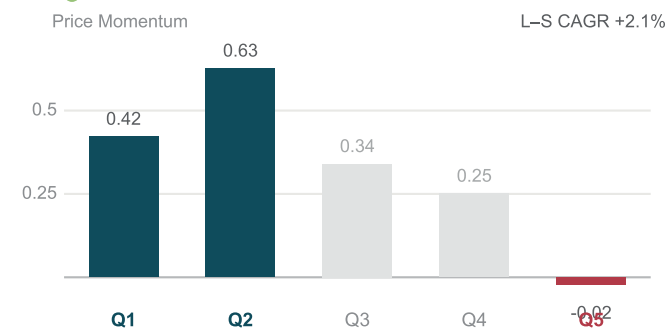
102 ● Retained Earnings to Total Assets



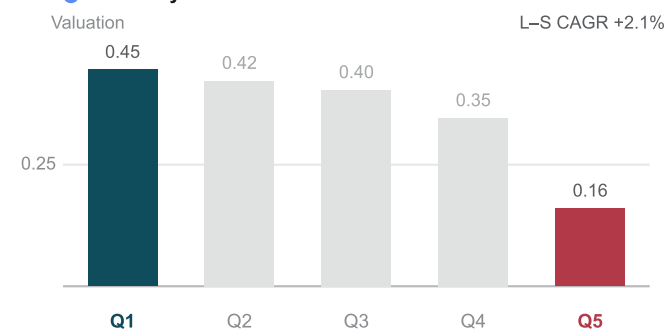
103 ● Pretax Profit Margin



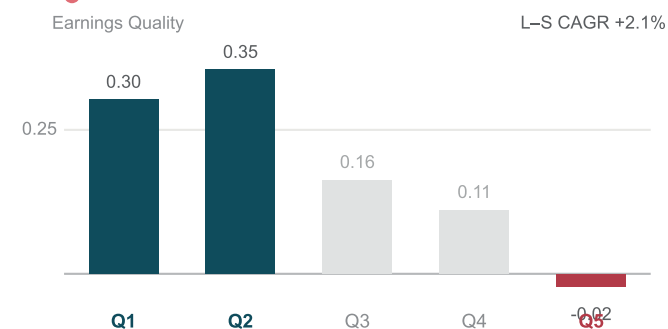
104 ● 26W Price Stat



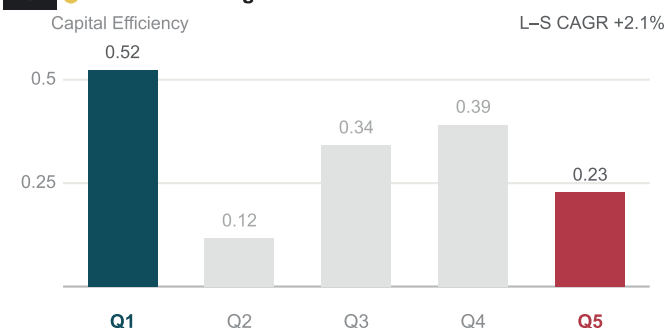
105 ● Solvency Ratio



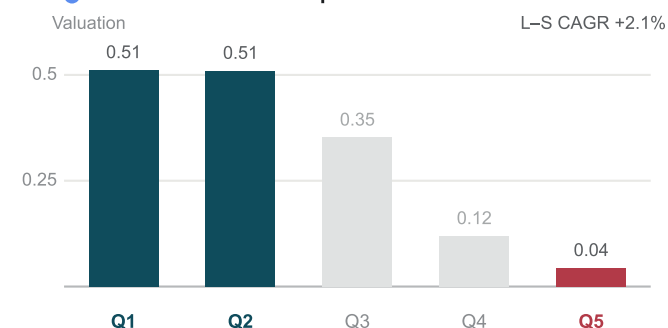
106 ● Accrual Ratio - Cash Flows

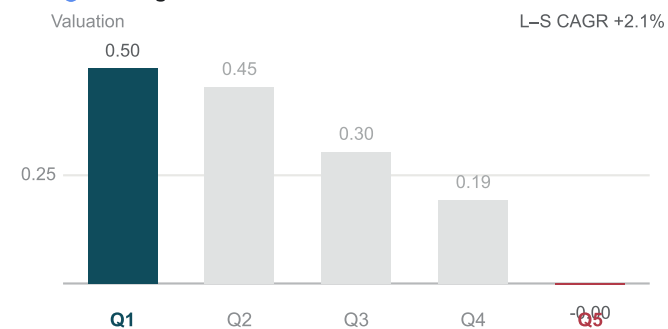
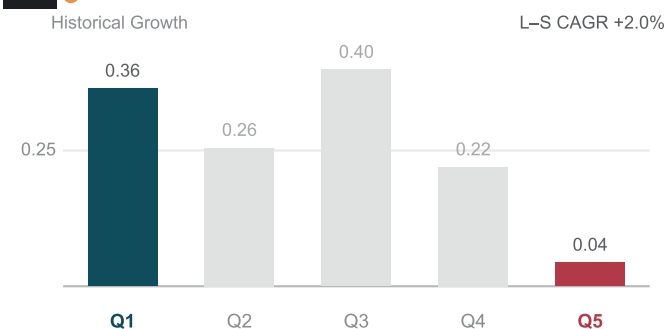
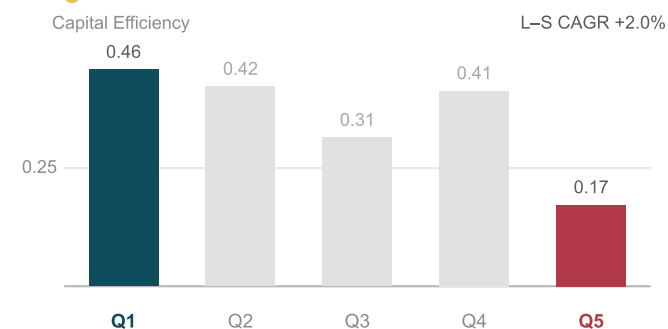
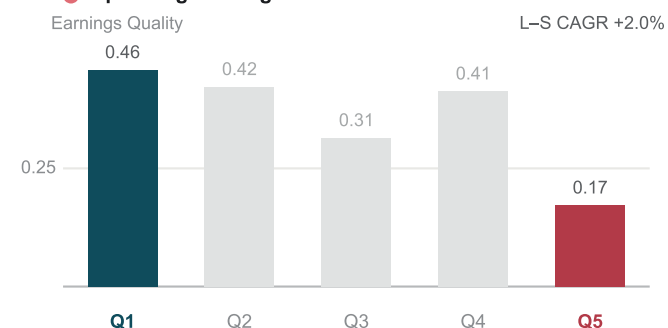
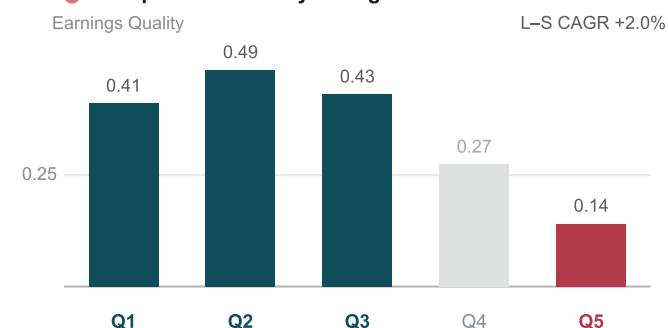


107 ● Interest Coverage Ratio

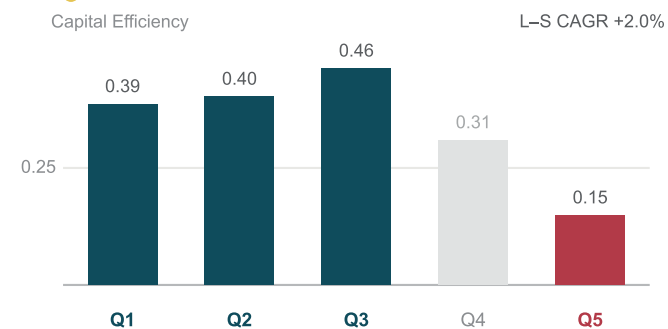


108 ● Net Cash Flow to Enterprise Value

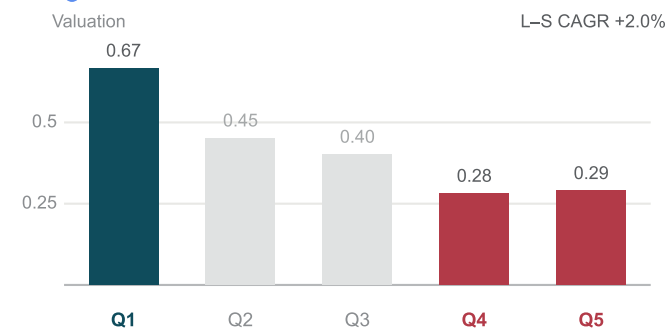


109 ● **Earnings to Price**110 ● **Adjusted Forward Earnings to Price**111 ● **Dividend Growth**112 ● **Return on Assets**113 ● **Operating Earnings to Assets Ratio**114 ● **Unexpected Inventory Change**

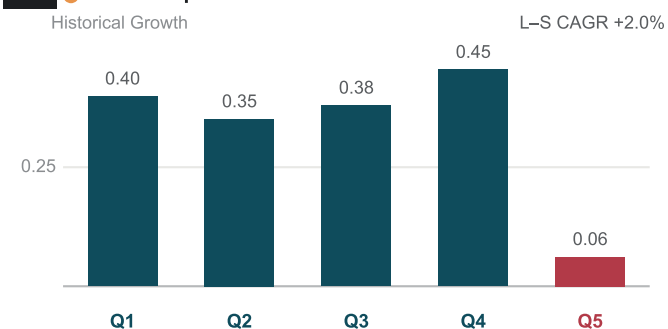
115 ● Debt to Assets Ratio



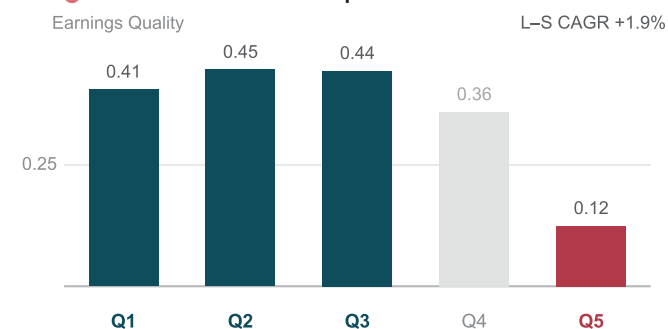
116 ● Forward Free Cash Flow to Price Ratio



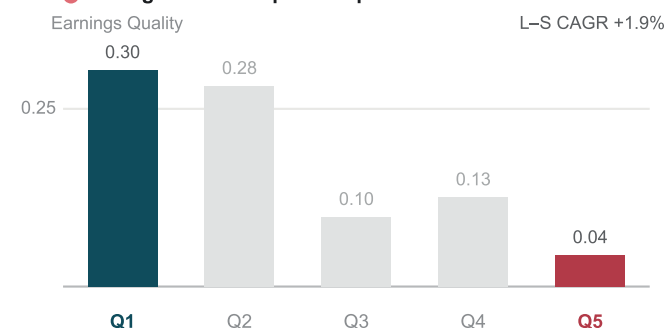
117 ● Dividend per Share Growth



118 ● Cash Flow to Invested Capital

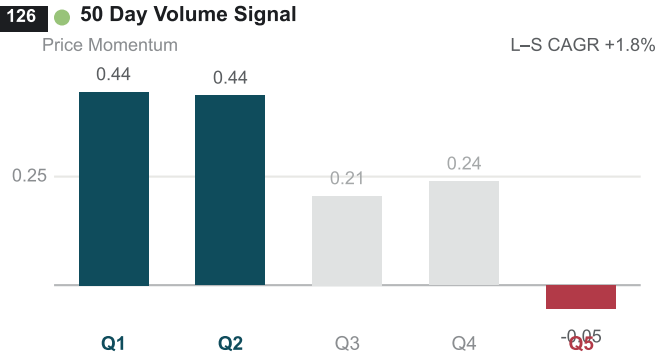
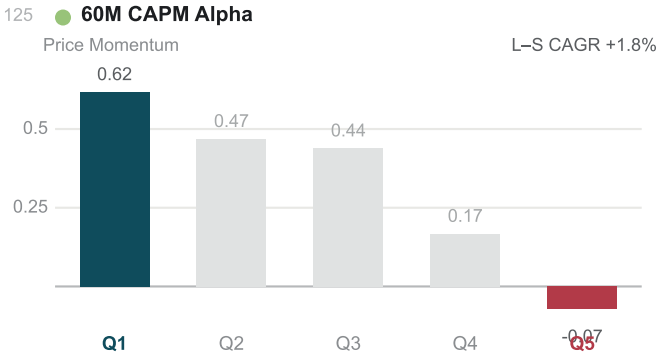
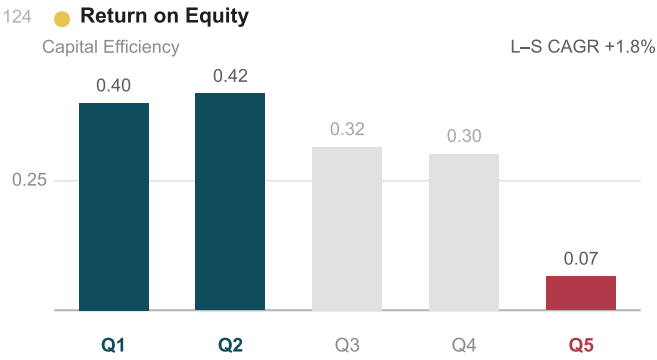
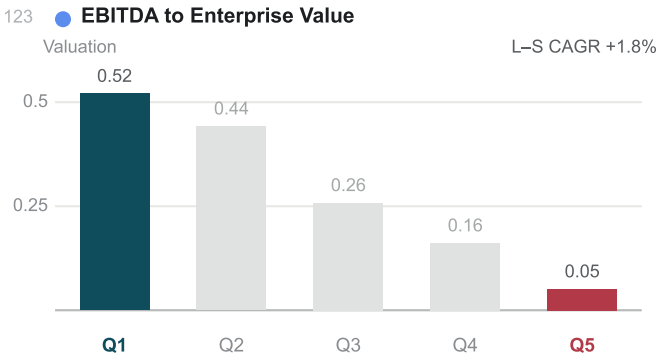
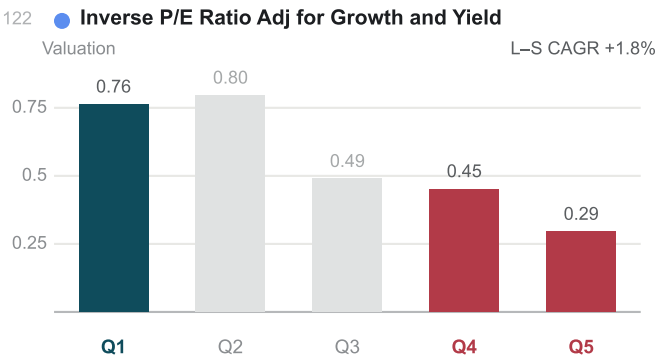
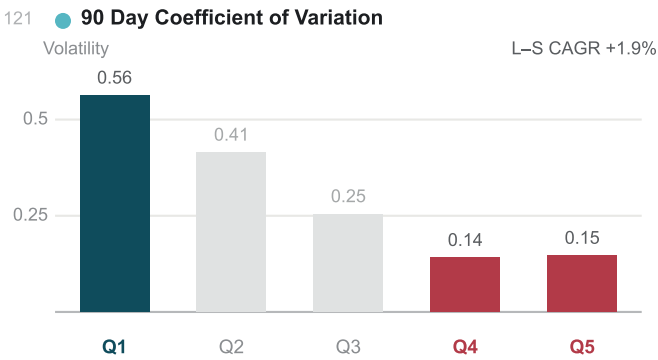


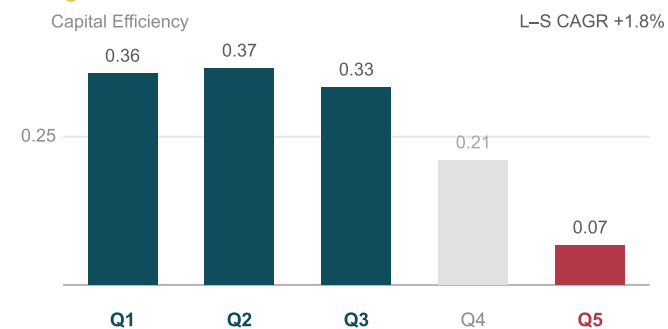
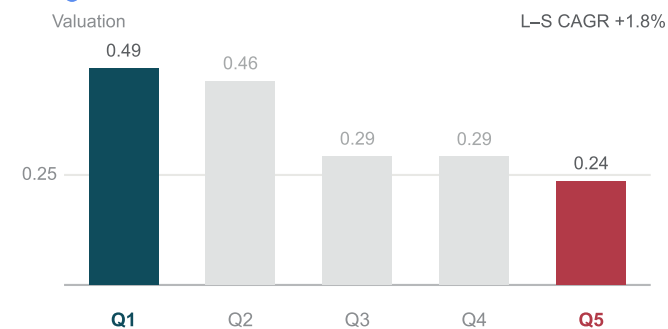
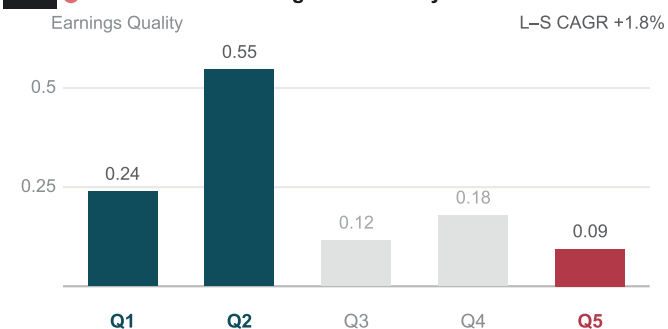
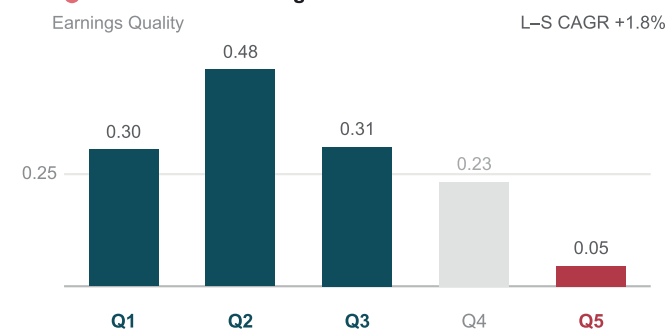
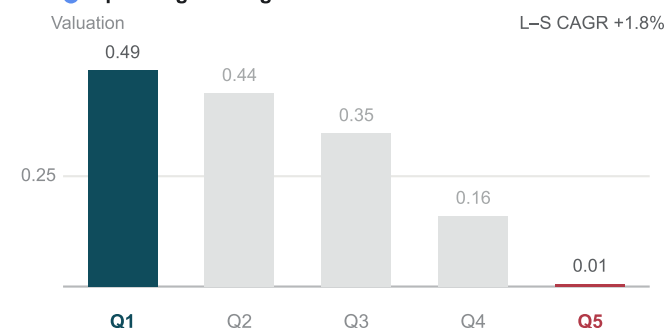
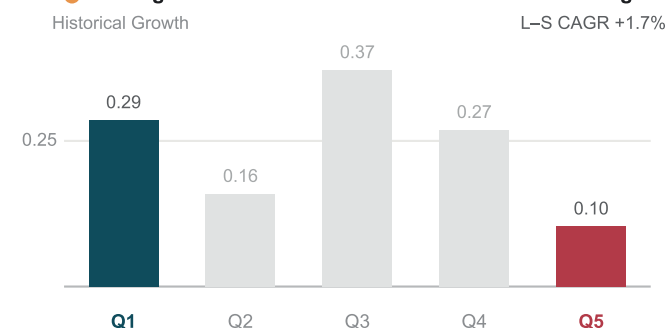
119 ● Change in TTM Depr. to Capex

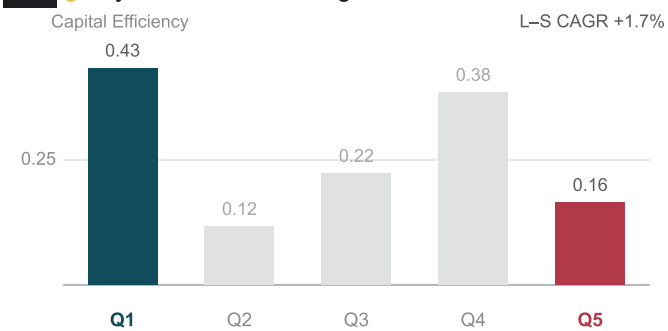
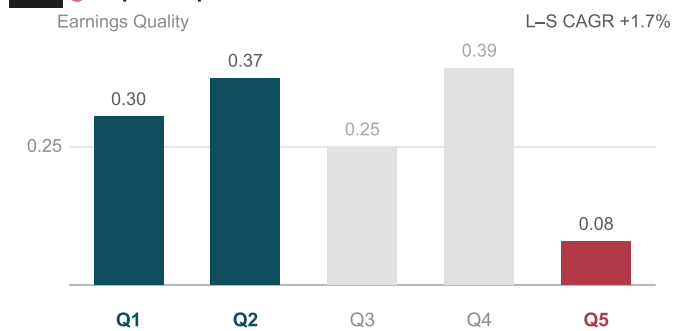
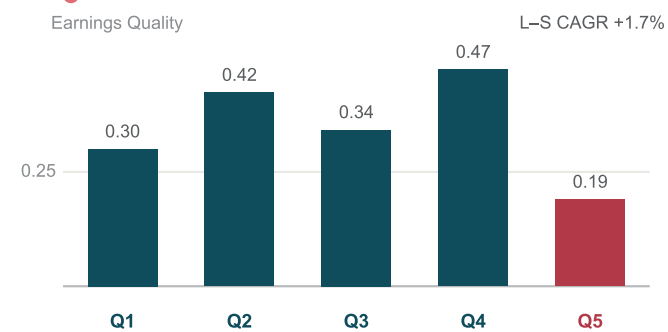
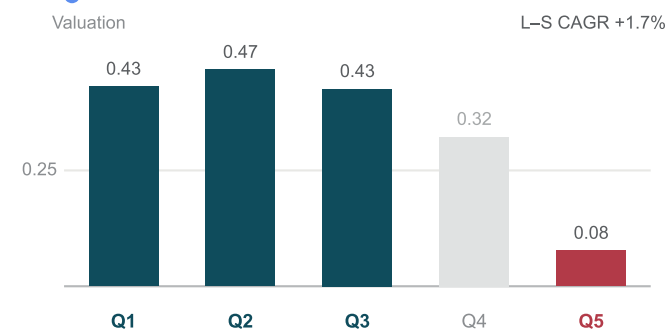
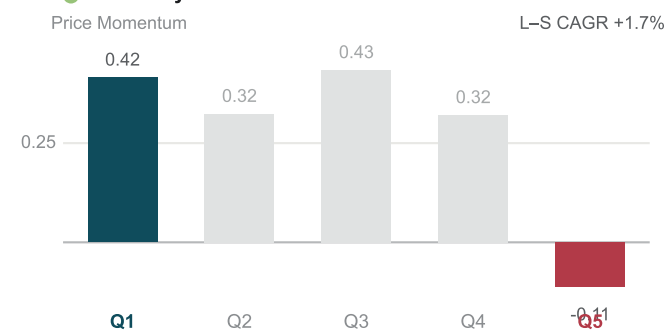


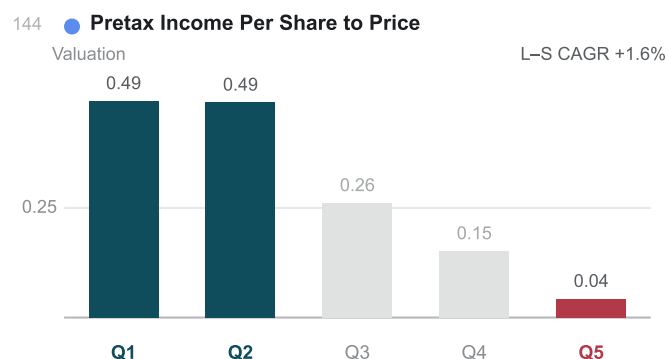
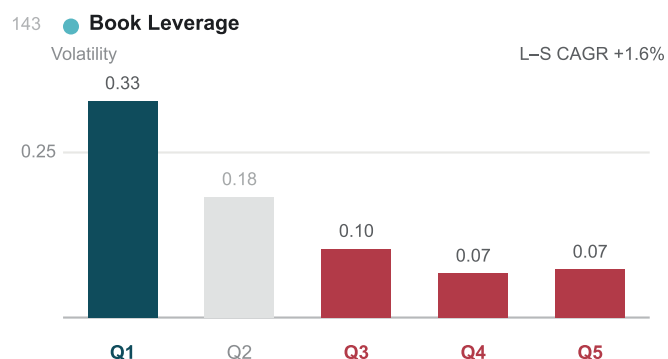
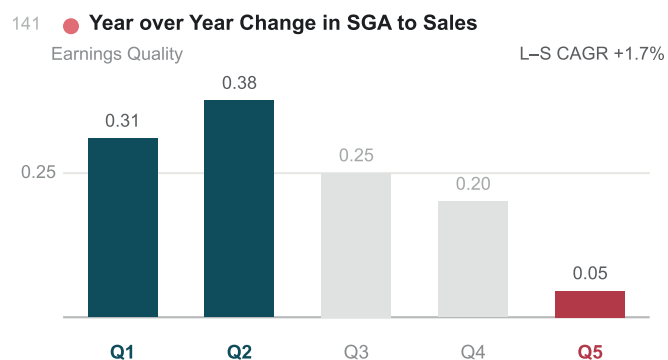
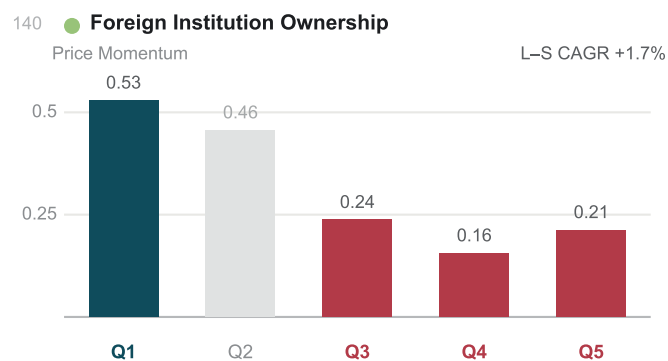
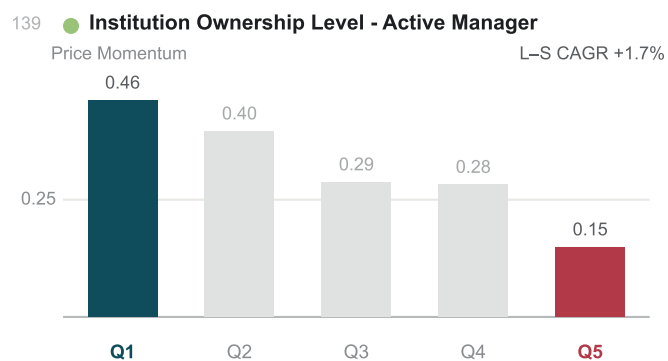
120 ● 3Y Chg in Cash Flow per Share



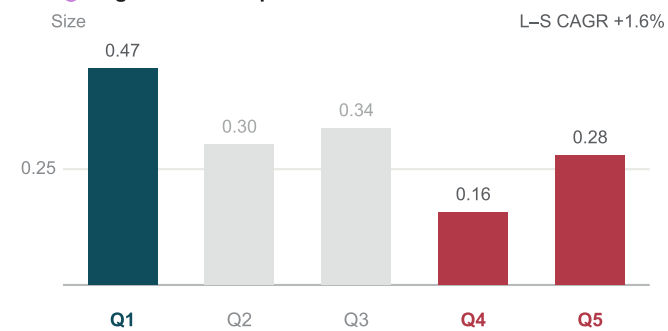


127 ● **EBITDA to Assets**128 ● **Net Current Assets to Price Ratio**129 ● **Year over Year Change of Inventory to Assets**130 ● **Year over Year Change of EPS to Sales**131 ● **Operating Earnings to Price Ratio**132 ● **Trailing 12-Month Receivables and Inventories/Trailing 12-M...**

133 ● Adjusted Interest Coverage Ratio**134 ● Capital Expenditure to Sales****135 ● Current Ratio****136 ● Normalized EP****137 ● Max Daily Return in the Past 6 Months****138 ● 3Y Chg in Free Cash Flow to Price**



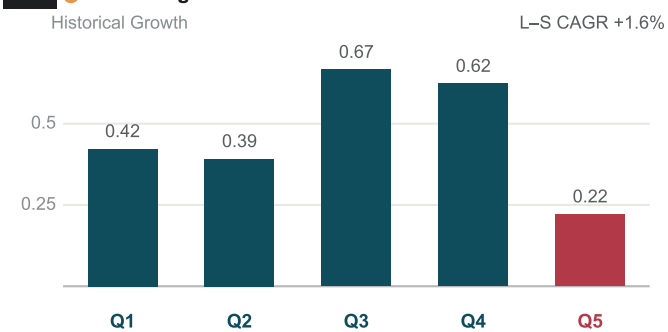
145 ● Log of Market Capitalization Cubed



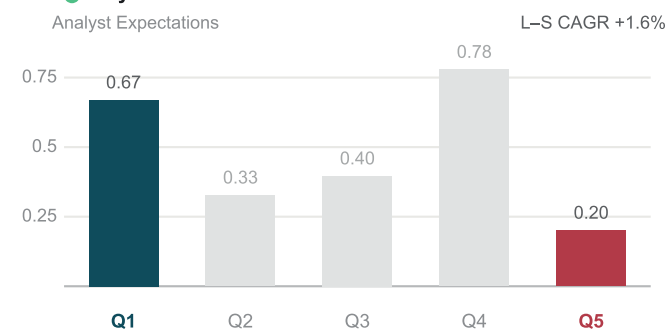
146 ● 3Y Chg in Asset Adjusted Free Cash Flow



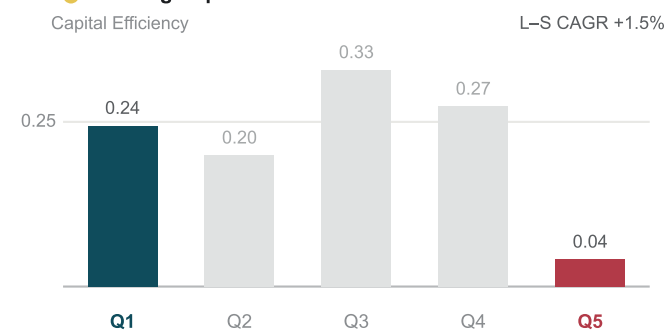
147 ● 3Y Average Annual Sales Growth



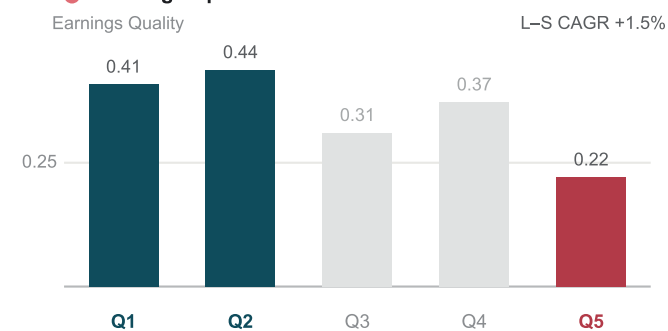
148 ● Buy-to-Sell Recommendation Ratio Less 3M SMA

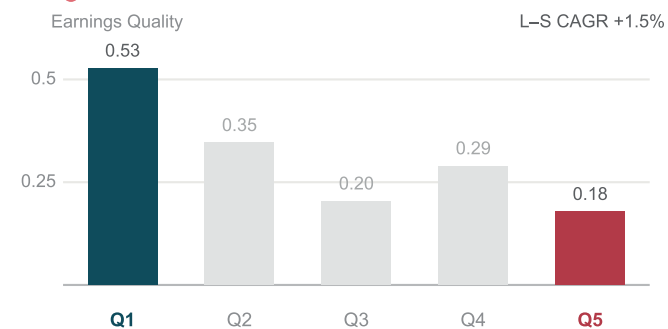
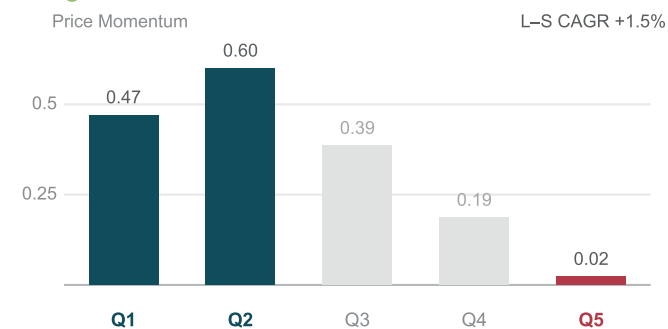
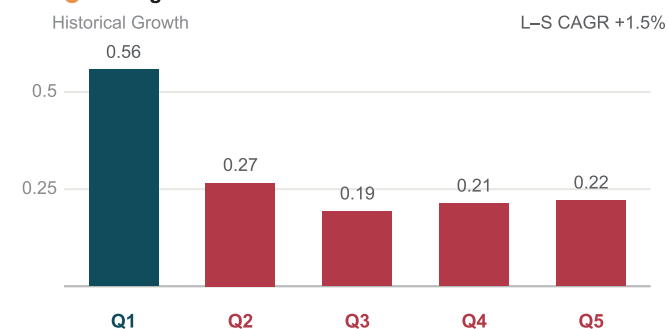
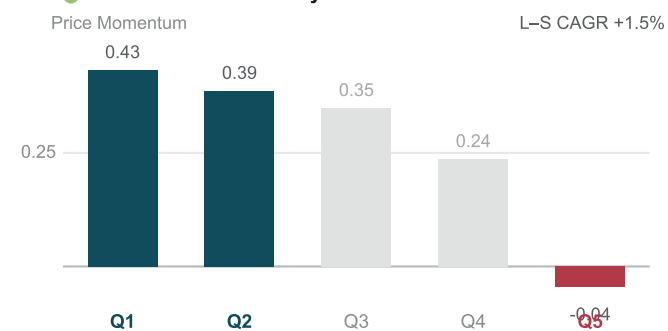
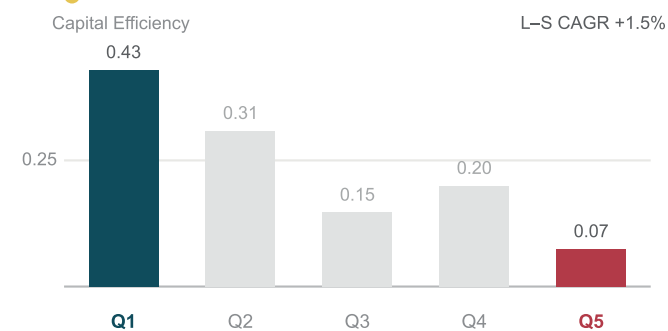


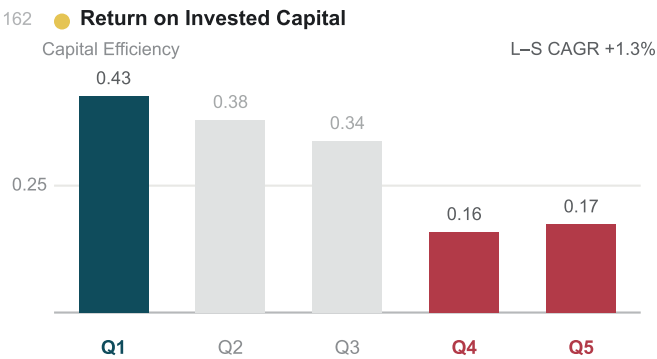
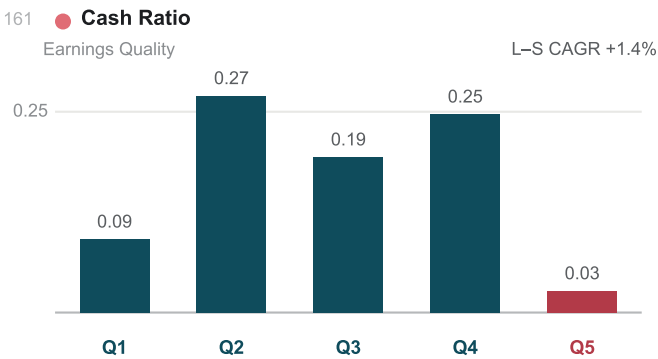
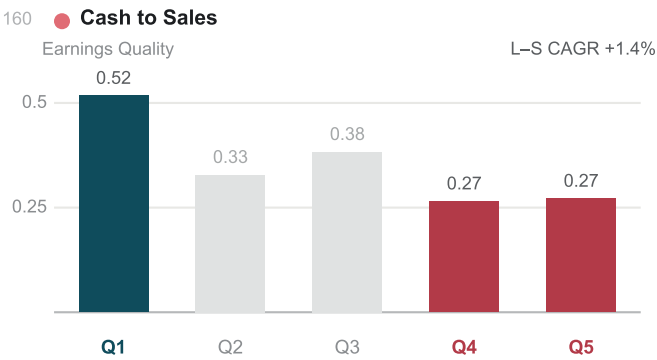
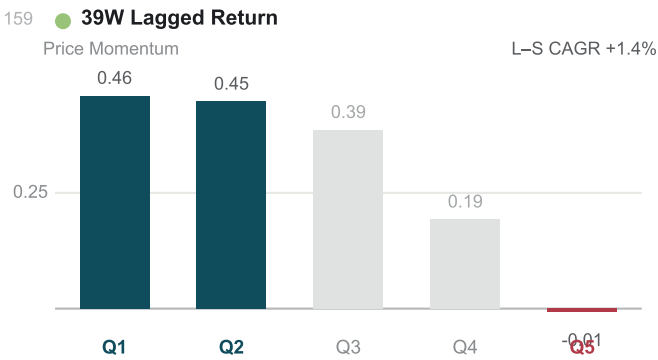
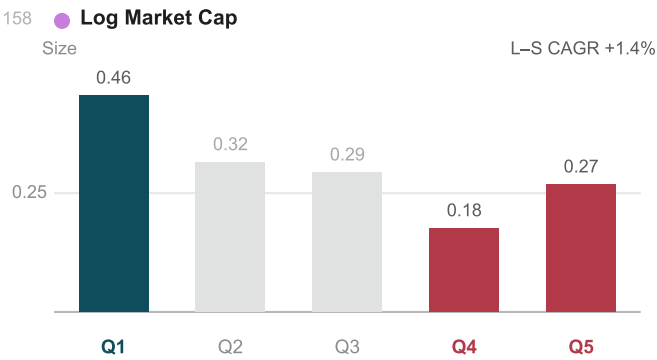
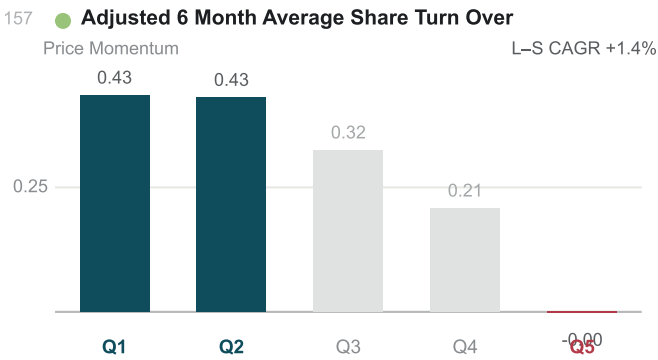
149 ● Working Capital Turnover Ratio



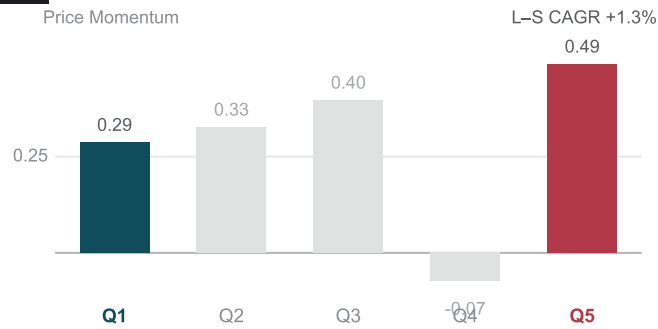
150 ● Working Capital to Total Assets



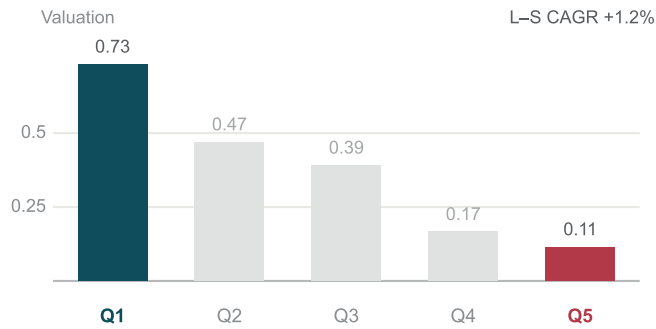
151 ● **Cash Burn Rate**152 ● **4W to 52W Price Oscillator**153 ● **Return on Net Tangible Assets**154 ● **3Y Chg in Cash Flow to Price**155 ● **Percent Above 260-Day Low**156 ● **Normalized ROE**



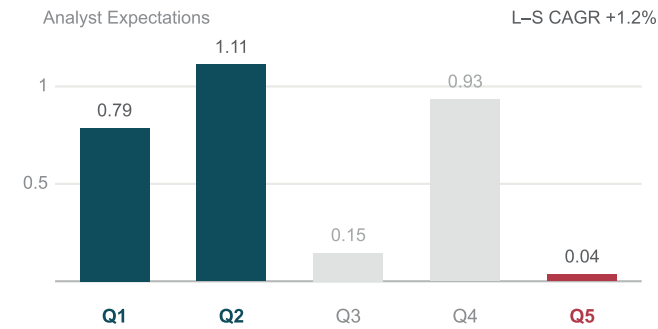
163 6 Month Average Share Turn Over



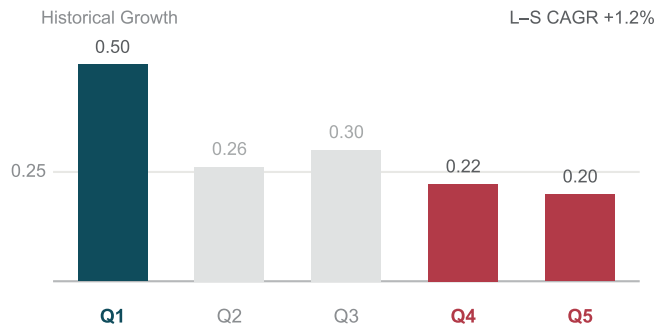
164 8-Week Change in 12-Month Forward Earnings Consensus Estima...



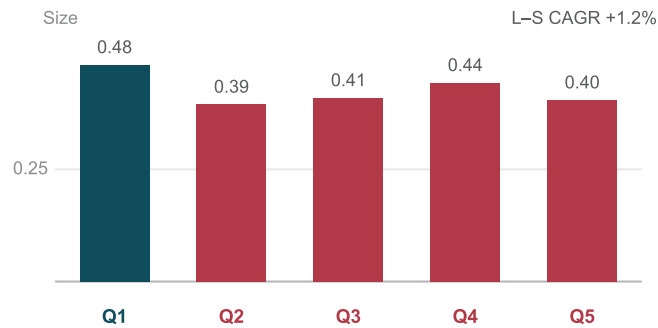
165 Adj Number of EPS FY2 Revisions



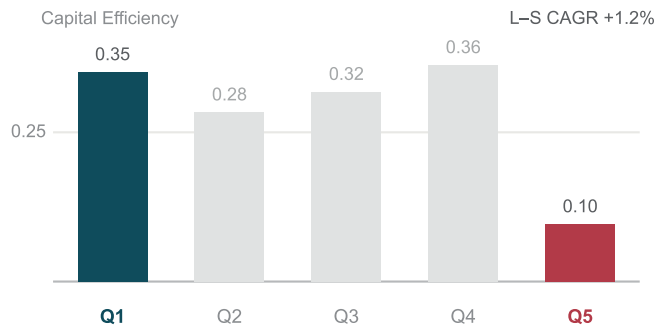
166 3Y Chg in Earnings to Price

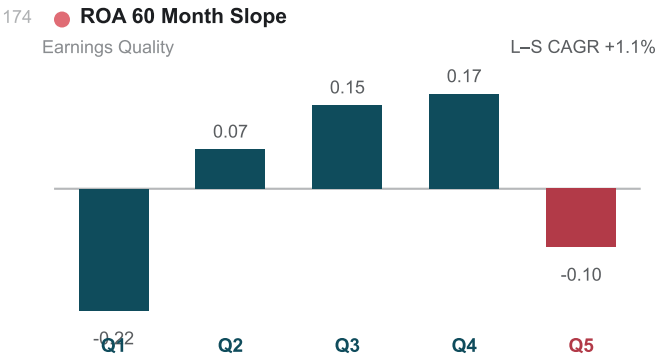
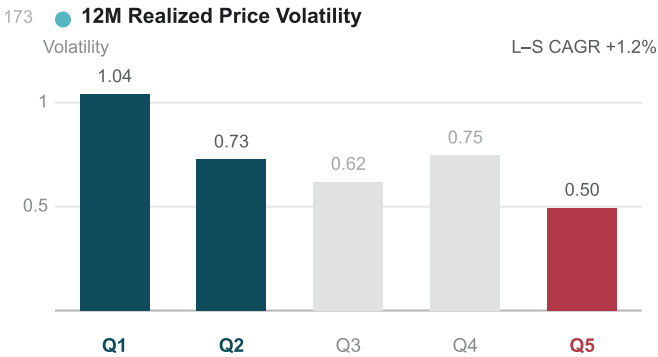
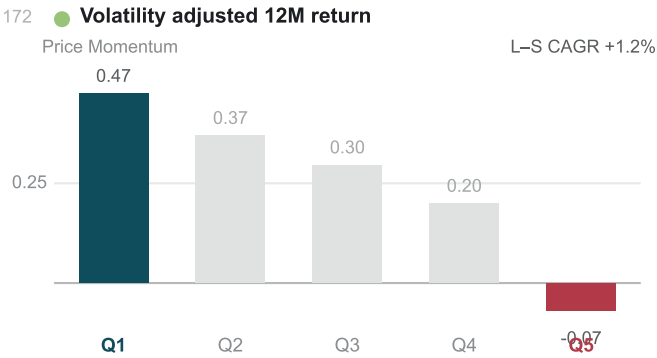
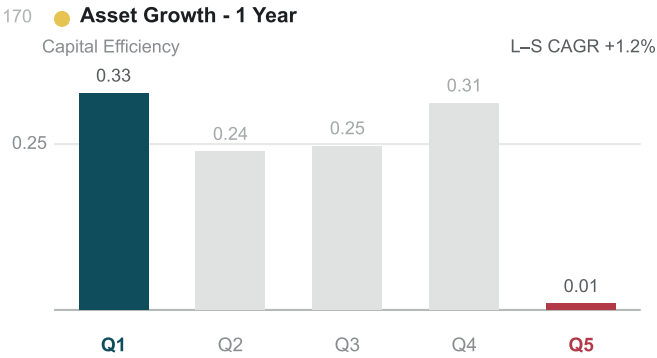
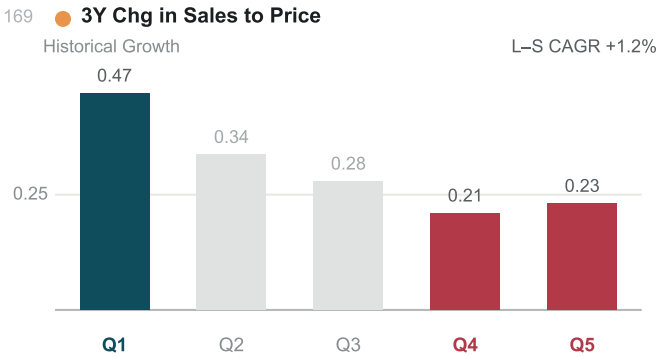


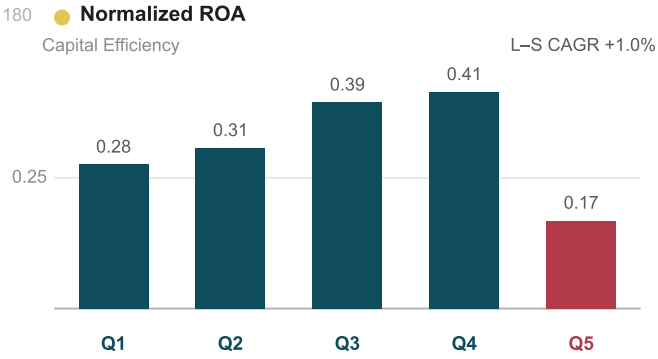
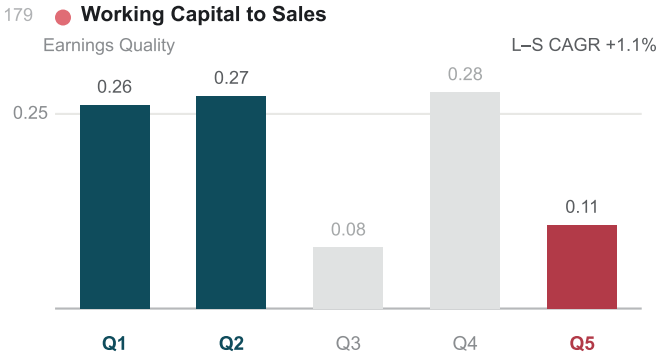
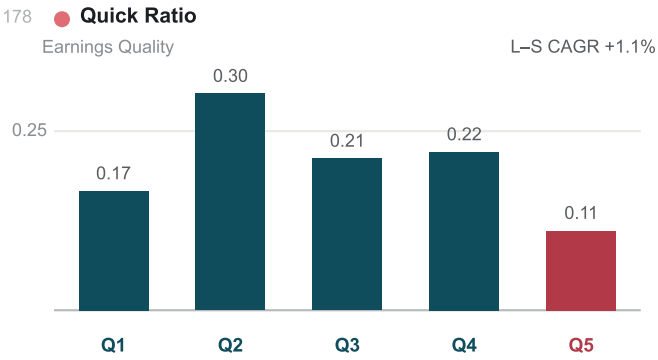
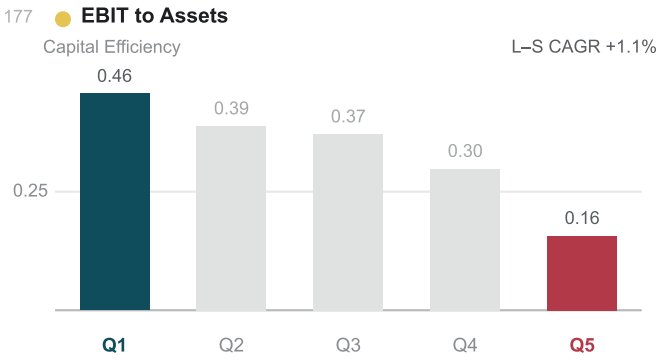
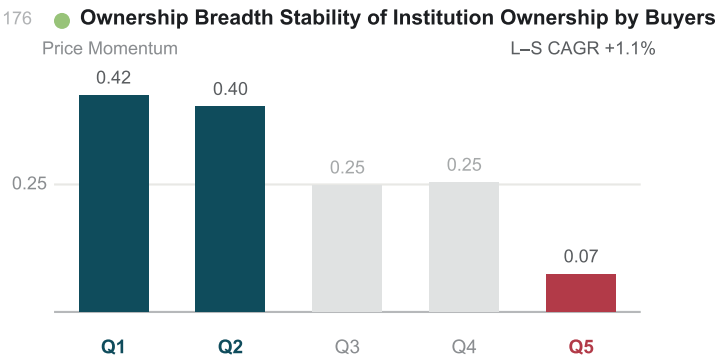
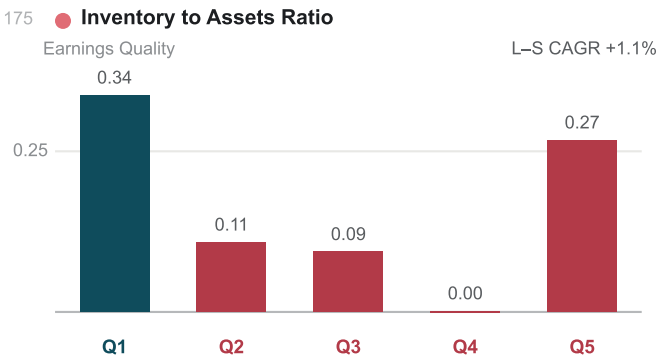
167 Log TTM Sales

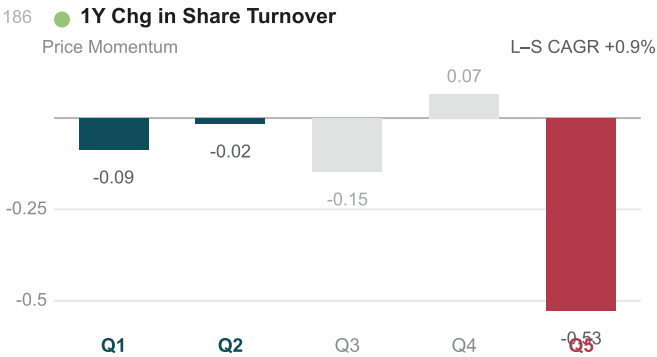
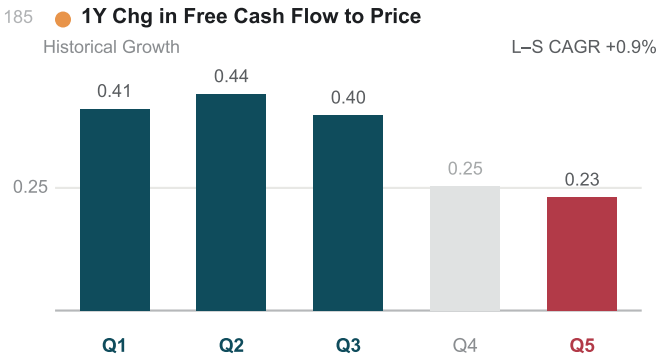
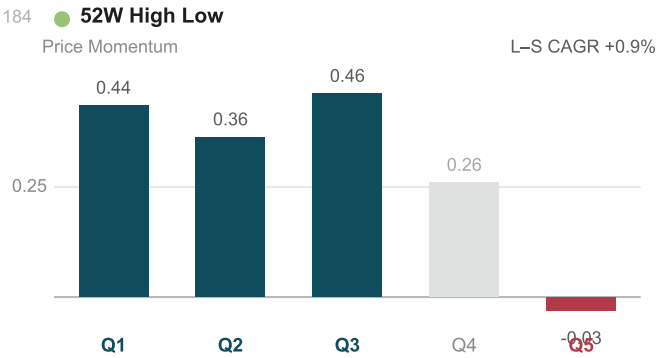
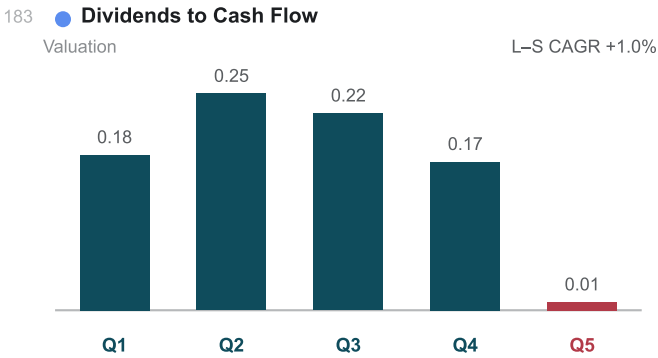
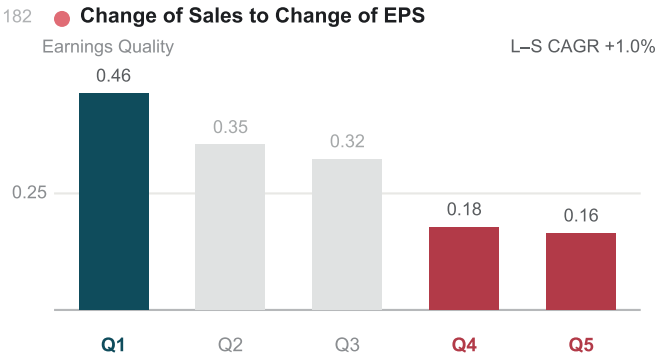
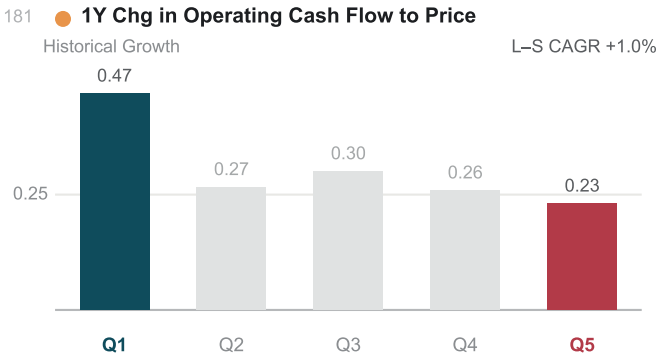


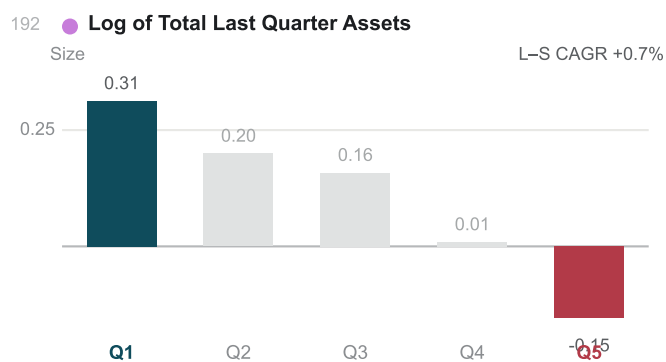
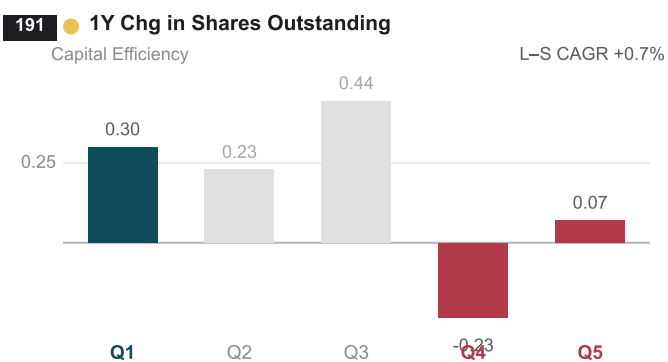
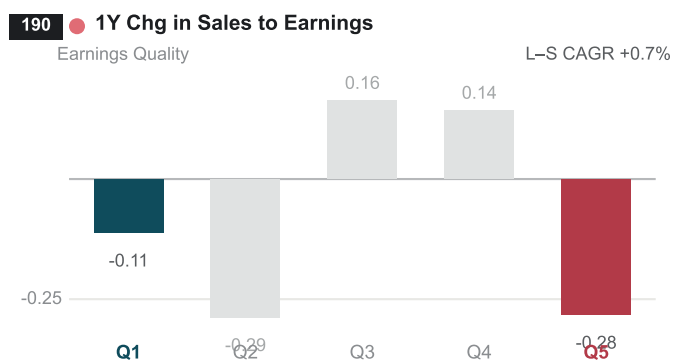
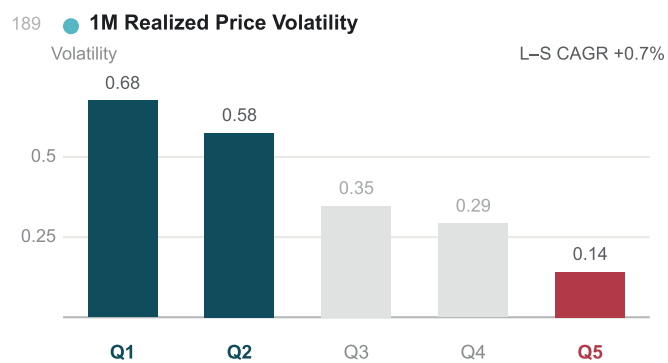
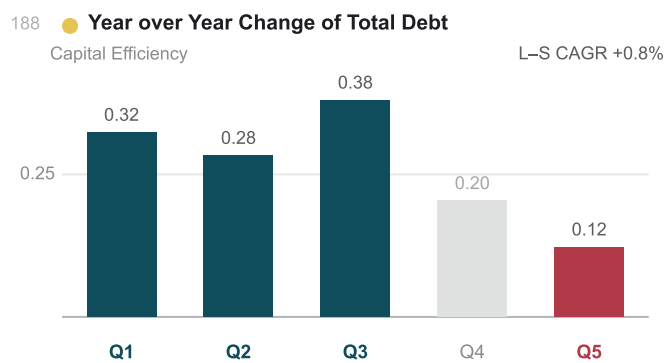
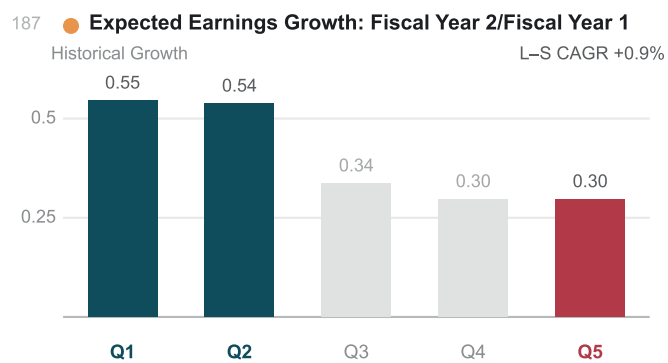
168 Degree of Financial Leverage



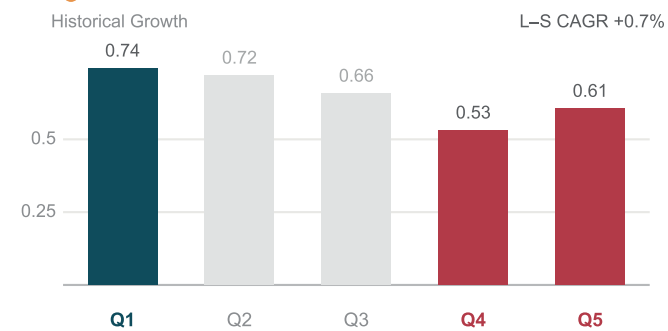




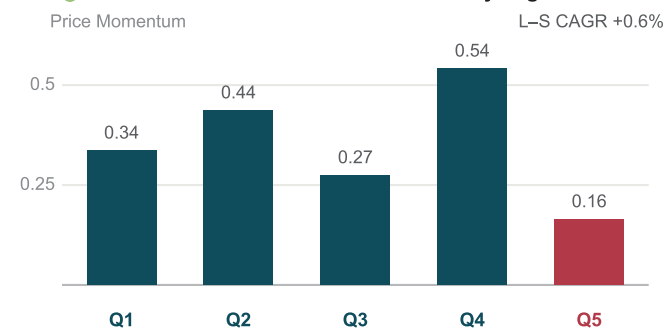




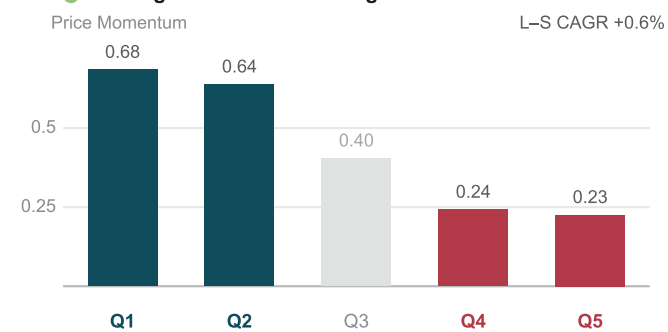
193 ● Sustainable Growth Rate



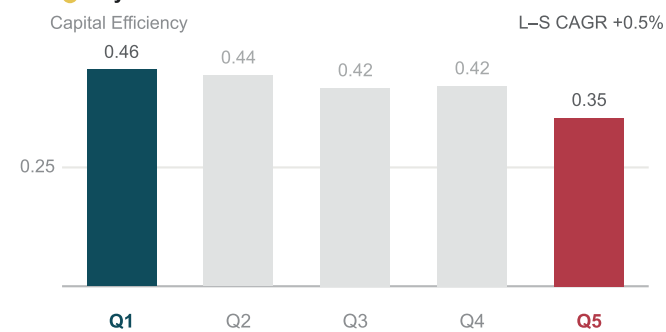
194 ● 52-Week Volume Price Trend with 20-Day Lag



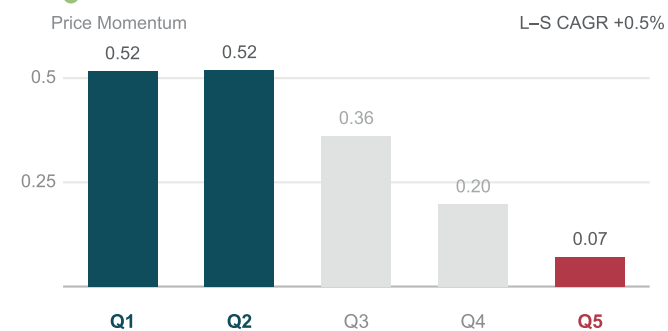
195 ● Closing Price to 52 Week High



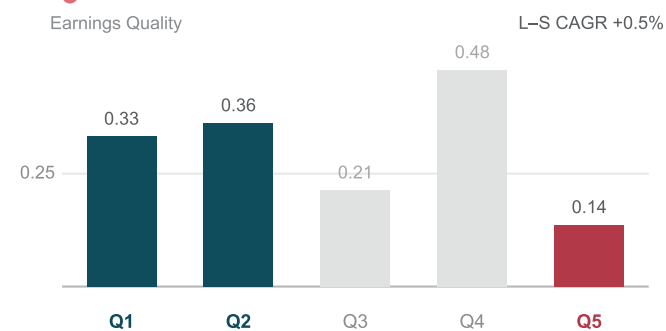
196 ● Payout Ratio

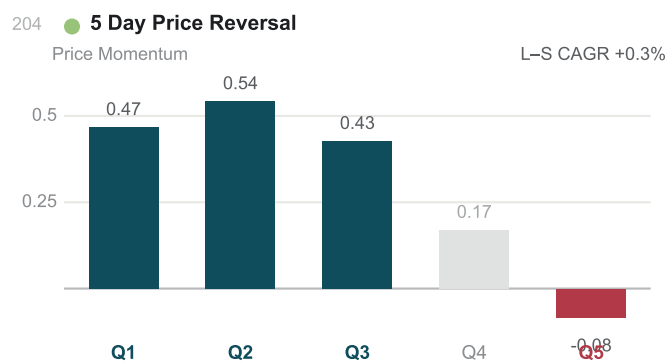
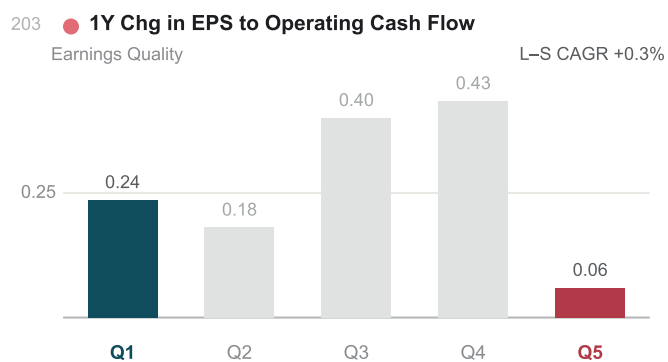
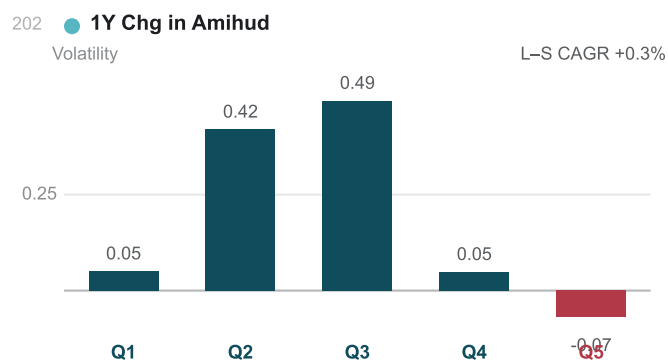
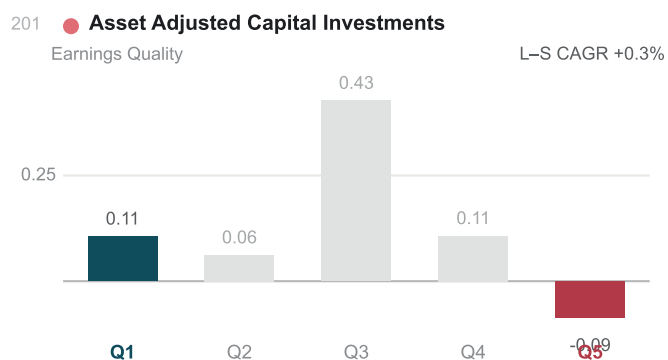
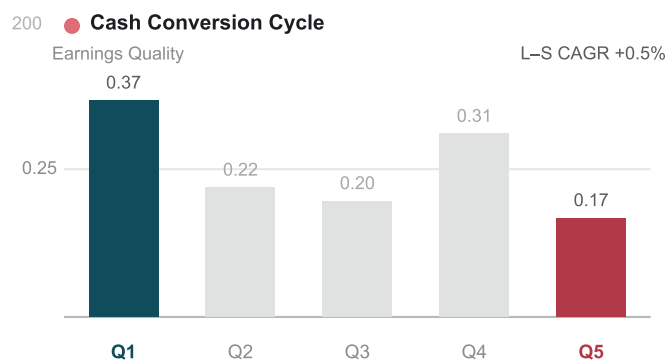
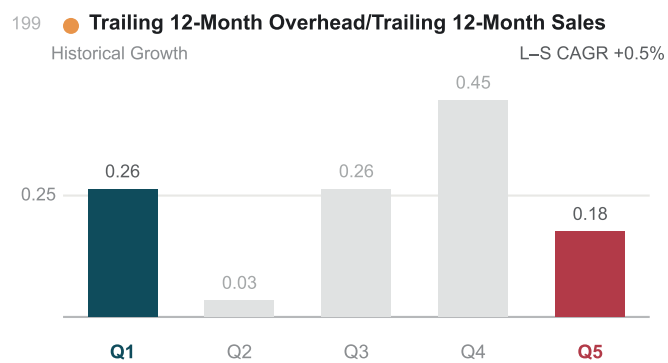


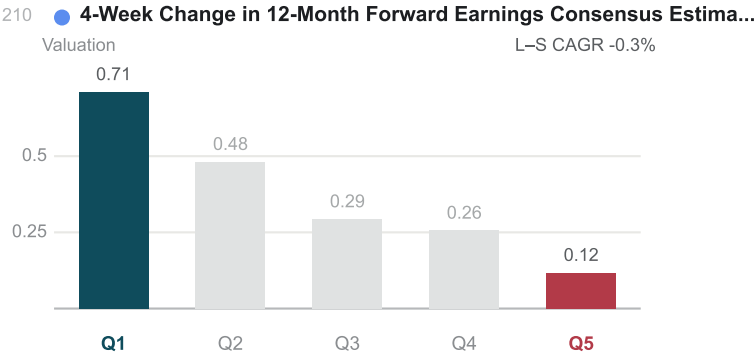
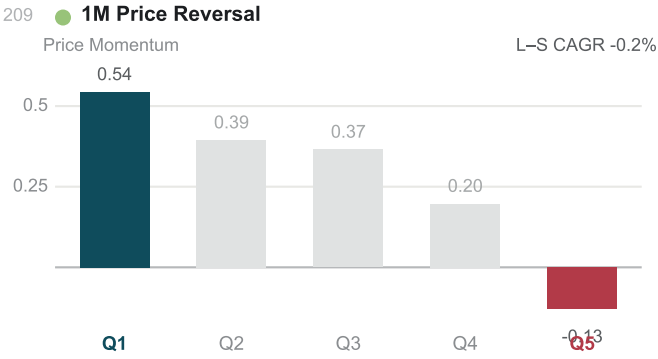
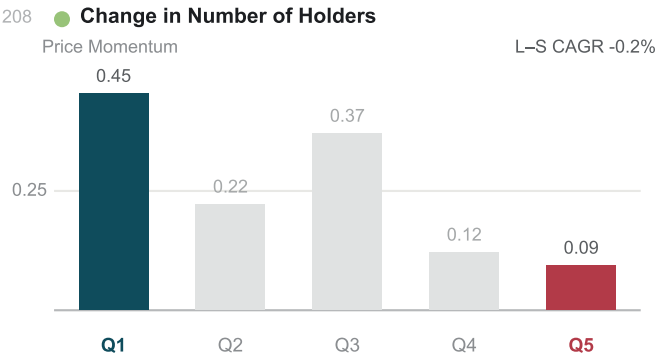
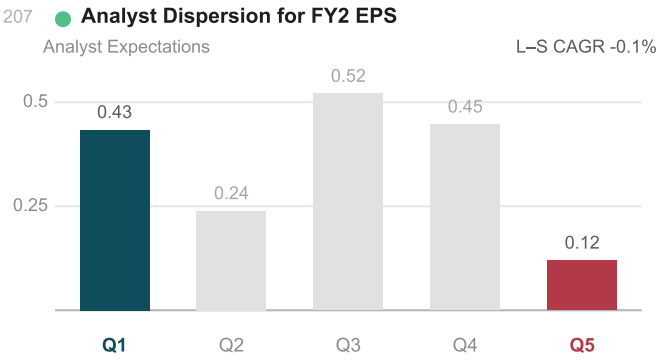
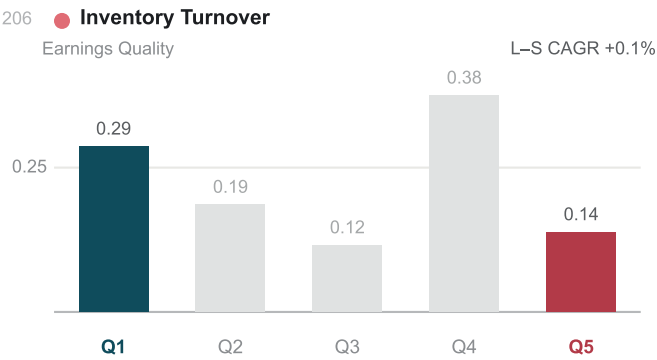
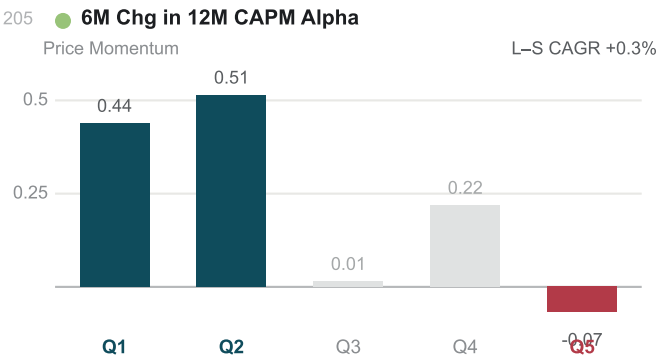
197 ● 9 Month Price Momentum

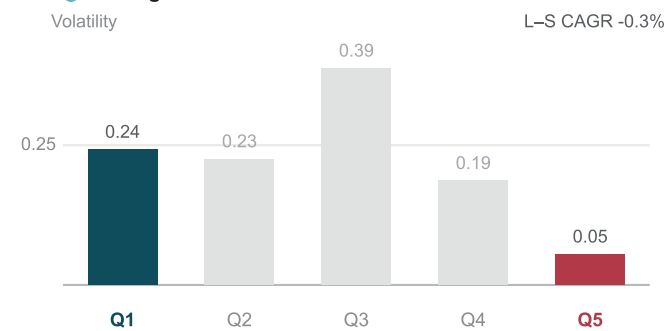
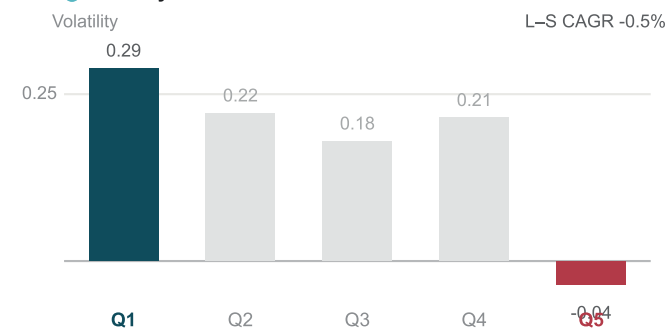
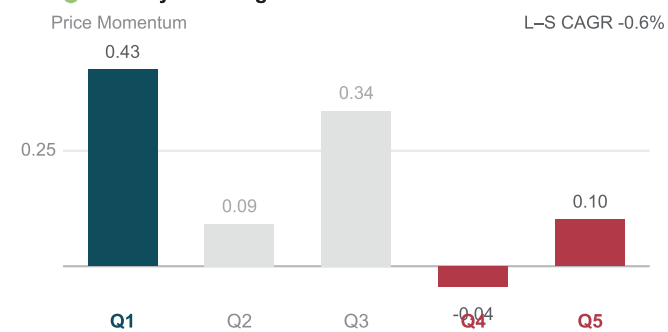
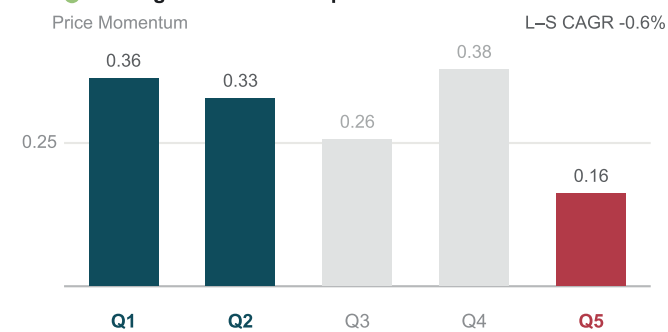
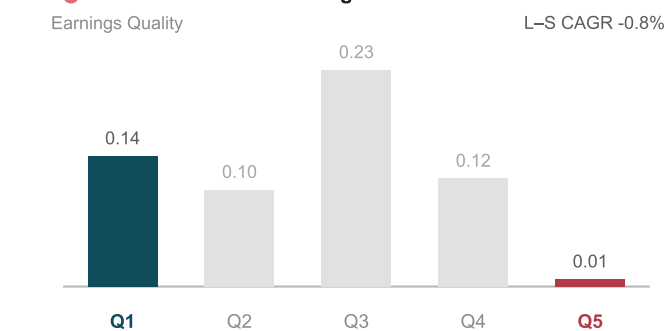


198 ● Balance Sheet Accruals







211 ● **1Y Chg in Debt to Assets**212 ● **130 Day Minimum Return**213 ● **Stability of Change in Number of Holders**214 ● **6M Chg in 36M CAPM Alpha**215 ● **Sales to Gross Profit Margin Growth**216 ● **1M Price High - 1M Price Low**